Direct Stochastic-Adversarial Equivalences in Online Learning

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Abstract

A thread of research – see, for example, [20] – has investigated a phenomenon in which the worst-case adversarial regret of certain online learning games can be evaluated (or upper bounded) by taking worst-case expectations of stochastic processes. One particularly direct and simple instance of this phenomenon was first observed by Thomas Cover in 1965 [4], in which we can completely characterize achievable data-dependent loss or risk profiles for a learner who is trying to sequentially predict the outcomes of a binary sequence. The similarities between Cover's work and the more complex equivalences derived from (e.g.) sequential Rademacher complexities have been noted – indeed, the authors of [7] credit Cover for inspiring their work – but the simplicity and directness of Cover's equivalence seemed to depend on the simplicity of the binary prediction setting.

We show that this is not so. We derive (relatively) simple, direct, and non-trivial equivalences in the setting of online linear optimization. This equivalence yields some results: starting from (stochastic) martingale inequalities, the equivalence to the adversarial learning setting allows us to show both:

(1) the tightness of the $\sqrt{T}$ external regret bound over the 2-dimensional Euclidean disk, and

(2) the optimality of the constant 2 in AdaGrad-style adaptive regret bounds over Euclidean balls

As part of this investigation in the OLO setting, we provide a sufficient condition (which we call spherical Lipschitzness), under which our equivalence holds, and under which we can guarantee that the constrained and unconstrained versions of the problem induce the same optimal strategy in the learner.

Beyond these specific contributions, we provide a unifying framework for a subset of these "stochastic-adversarial equivalences," which allow for direct analysis of regret bounds via expectations of the comparator term under stationary processes. We define these equivalences in a general sense, and give conditions under which learning problems will enjoy a direct equivalence. We explore these conditions in the settings of binary prediction and online linear optimization. In the binary prediction setting, we derive new "Cover-like" results for a variety of losses.

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Chapter 1

Introduction

At the core of the theory of online learning is the value of learning games. Given an online learning setting, the value measures the best possible loss (with respect to some benchmark) that a learner can achieve, regardless of the data the learner is presented with. It is often easiest to think of these settings as a game between the learner and an adversarial Nature, in which Nature is actively trying to undermine the learner's performance with respect to the benchmark. The learner's ultimate performance relative to the benchmark, assuming that both the learner and Nature are playing optimally, is called the value.

Many researchers (e.g. [4], [12], [22]) have noticed that the value of learning games can often be upper-bounded, or even evaluated exactly, by looking at expectations of particular functions over either i.i.d. proceeses or martingale difference sequences. This non-constructive approach allows for the learnability of some problems to be assessed, without needing to specify an optimal algorithm tailored to the learning setting – that is, without needing to specify optimal strategies for the learner and Nature. It also has the advantage of opening a kind of dialogue between the theory of adversarial learning and the (math underlying) the theory of stochastic learning. Worst-case results from stochastic or empirical processes can inform analysis of the value of online learning games, and as [20] showed, the converse can be true as well.

This thesis considers a class of these "stochastic-adversarial equivalences," in which the value of the adversarial learning game is exactly equal to the expected value of the learner's optimal performance, assuming that Nature is not fully adversarial, but rather is forced to pre-commit to a

restricted stochastic process.

This may not yet be fully clear, but hopefully some more technical specificity can help elucidate. Throughout this thesis, we will analyze the online adversarial learning scenario with no side information: an online learner makes sequential predictions chosen from a report set $\mathcal{A}$ against environment outcomes chosen from an outcome set $\mathcal{Y}$. The learner is evaluated on each of T rounds by a loss function $\ell : \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$. ($T$ is fixed ahead of time.) We'll write the learner's algorithm as a function $\text{Alg} : \mathcal{Y}^* \to \mathcal{A}$ which generates reports from a history of outcomes.

The value of such games is given by the worst-case loss of learner's algorithm across outcome sequences $\mathbf{y}_{1:T} \in \mathcal{Y}^T$, offset by a comparator benchmark. We'll consider a somewhat general form of benchmark, which allows for any arbitrary dependency on outcomes, given by a function $\phi : \mathcal{Y}^T \to \mathbb{R}$. This accommodates, for example, standard external regret, taking $\phi(\mathbf{y}_{1:T}) = -\inf_{a' \in \mathcal{A}} \sum_{t=1}^{T} \ell(a', \mathbf{y}_t)$. The value of the learning game under this general form of regret is then: $\inf_{\text{Alg}} \sup_{\mathbf{y}_{1:T}} \sum_{t=1}^{T} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T})$.[1]

We contrast the adversarial setting, which makes no assumptions about how the data are generated, against the analysis of regret under the assumption that data are generated by a stochastic process $Y_{1:T}$ with joint distribution $P_{1:T} \in \Delta(\mathcal{Y}^T)$, and that the learner is Bayes optimal. That is, the learner suffers the per-round Bayes Risk $L(P_t \mid \mathbf{y}_{1:t-1}) = \inf_{a \in \mathcal{A}} \mathbb{E}_{P_t | \mathbf{y}_{1:t-1}} \ell(a, Y)$, where P_t is the t-th conditional distribution of the joint distribution $P_{1:T}$.

1.1 Direct Stochastic-Adversarial Equivalences With No Distribution Restrictions

Begin by placing no restrictions on the distribution $P_{1:T}$. That is, $P_{1:T}$ can be any arbitrary joint distribution over $\mathcal{Y}^T$. Then, if we make some convexity and compactness assumptions about the learning task, a generalization of Theorem 1 from [1] tells us that our minimax regret guarantees

[1] Most other literature has ϕ subtracted from the loss instead of added. I found that this made some of the proofs and a few statements awkward, with lots of minus signs, so throughout the thesis, ϕ will be added to the loss.

are identical, whether or not we assume the data are generated by a stochastic process:

$$\inf_{\text{Alg}} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = \sup_{P_{1:T} \in \Delta(\mathcal{Y}^T)} \mathbb{E}_{P_{1:T}} \left[\sum_{t=1}^{T} L(P_t \mid Y_{1:t-1}) + \phi(Y_{1:T}) \right] \tag{1.1}$$

This is a somewhat surprising result. It doesn't matter whether an adversarial Nature plays deterministically at each round after observing the learner's play, or must commit at the start of the game to following a specific stochastic process, which the learner knows and against which the learner can optimize. The learning task is equally difficult in both framings.

Of course, it's generally difficult to evaluate the right-hand-side of eqn. (1.1), or even find a nice upper bound on it. Inequalities from stochastic processes typically assume some structure on the distribution.

So, it would be nice if we could restrict the allowable data-generating processes to a more structured class of distributions $\mathcal{P}_{1:T} \subset \Delta(\mathcal{Y}^T)$. It turns out we often can. These restrictions are the subject of the rest of the thesis. We'll consider these restrictions generally and give abstract conditions under which a "stochastic-adversarial" equivalence holds. Then, we'll apply these conditions in learning settings. In these applied settings, we'll be considering $\mathcal{P}_{1:T}$ which consist of i.i.d. processes or martingale difference sequences.

We can observe that in the case where $\mathcal{P}_{1:T} = \{P^T : P \in \Delta(\mathcal{Y})\}$ (the set of i.i.d. processes), then the right-hand-side of eqn. (1.1) is the minimax regret in the standard statistical learning setting. This leads to an interesting interpretation: in settings which admit an equivalence like eqn. (1.1) under i.i.d. $\mathcal{P}$, we might as well assume Nature is adversarial. That is, we can get the robustness of adversarial algorithms without increasing worst-case regret in a world that is actually statistical. But before we get too excited about this apparent "best-of-both-worlds", it's worth reflecting on the ways in which the statistical learning setting *already* embeds some hidden adversarial structure. In standard statistical (PAC) learning, a statistical learner must succeed on every possible i.i.d. distribution, even those which are chosen precisely by an adversary. A good example to keep in mind is a learner who must predict the outcome of a coin flip. (We allow the coin to be biased, but

the coin must remain the same across each timestep). For any coin with even the slightest bias, the learner will eventually be able to detect and exploit the bias, and will eventually be right more often than wrong. And in an imprecise sense, "almost all" coins will have some slight bias and will be learnable. However, a pure statistical learner which can predict meaningfully on almost all coins is inadequate: we need it to learn all coins, including the perfectly fair coin. But given this perfectly fair coin, no learner can be right more often than wrong in the long run. It is the precise, adversarial selection of zero bias which makes coin prediction a difficult problem in the limit. This phenomenon has been noted before in the literature, and the next section will explore it in more technical detail.

1.2 Direct Equivalences in Prediction of Binary Sequences Under i.i.d. Restrictions

The first direct equivalence (in the sense of eqn. (1.1)) in the literature goes back to [4]. There, Cover investigated randomized predictions of binary sequences evaluated under expected 0-1 loss. In our notation, $\mathcal{Y} = \{0, 1\}$, $\mathcal{A} = [0, 1]$, $\ell(p, y) = (1 - p)y + p(1 - y)$, and $\mathcal{P}_{1:T} = \{\text{Ber}(1/2)^T\}$, consisting of the unique hardest i.i.d. distribution to learn. Cover's result requires $\phi : \mathcal{Y}^T \to \mathbb{R}$ to satisfy two conditions. First, it must satisfy a tail stability property (property 4.6 in [4]), for all $t \leq T$ and all starting outcomes $\mathbf{y}_{1:t-1} \in \{0, 1\}^{t-1}$:

$$\left| \mathbb{E}_{\text{Ber}(1/2)^{T-t}} \left[\phi(\mathbf{y}_{1:t-1}, 1, Y_{t+1:T}) \right] - \mathbb{E}_{\text{Ber}(1/2)^{T-t}} \left[\phi(\mathbf{y}_{1:t-1}, 0, Y_{t+1:T}) \right] \right| \leq 1 \tag{1.2}$$

(Note that this condition is essentially a Lipschitz condition.) Second, ϕ must satisfy:

$$\mathbb{E}_{\text{Ber}(1/2)^T} [\phi(Y_{1:T})] = T/2 \tag{1.3}$$

For any such ϕ satisfying both eqn. (1.2) and eqn. (1.3), Cover tells us that:

$$\inf_{\text{Alg}} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) - \phi(\mathbf{y}) = 0 = \left(\sum_{t=1}^{T} L(\text{Ber}(1/2)) \right) + \mathbb{E}_{\text{Ber}(1/2)^T} \left[-\phi(Y_{1:T}) \right] \tag{1.4}$$

That is, we have a direct characterization of adversarial minimax "regret", across a wide variety of possible formulations of regret defined by ϕ, simply by taking the expectation of ϕ under a stationary stochastic process.

[21] provide a tutorial on the use of Cover's result. They show how to inject "prior knowledge" of a binary prediction problem by choosing a particular ϕ. They also show how one can apply Cover's result, evaluating an expectation to get the conditional value of the game, and then using this conditional value to construct an optimal algorithm via a "relaxation" method. Finally, they extend Cover's result to the multiclass setting and the supervised setting (adding side information), under a slightly stronger stability condition.

Later, we will extend Cover's result in a different direction, varying the learner's report space and loss function.

1.3 Direct Equivalences in Unconstrained Online Linear Optimization Under Martingale Restrictions

[12] apply the above minimax theorem to the problem of unconstrained online linear optimization. In our notation, they take $\mathcal{Y} \subset \mathbb{R}^d$ to be any d-dimensional polytope, $\mathcal{A} = \mathbb{R}^d$, $\ell(a, y) = \langle a, y \rangle$, and $\mathcal{P}$ to be the set $\mathcal{P}^{(0)}$ of all distributions with zero mean (so that $\mathcal{P}_{1:T} = \mathcal{P}^{(0)}_{1:T}$ is the set of all martingale difference sequences). They show that $\mathcal{P}^{(0)}$ is a valid restriction, assuming only the boundedness of ϕ, and go on to consider a variety of algorithms based on the value of the game. Interestingly, both this result and Cover's result were accompanied by discussions of betting strategies.[2] Later, we will give sufficient conditions on ϕ for direct equivalence to hold in the OLO setting with constrained learners.

1.4 Indirect Equivalences in Broad Classes of Learning Settings

The minimax result (1.1) also helped inform subsequent work towards what one might call indirect equivalences. Such equivalences do still relate the value of adversarial learning games to

[2] We'll return to this observation in Section 6.1.

expectations of stochastic processes, but the stochastic process which bounds the game value is not the same process as in eqn. (1.1). One key type of indirect equivalence comes via the notion of sequential Rademacher complexity, introduced by [22]. Such equivalences use a symmetrization approach to transform the supremum over the full probability space to a different supremum over Rademacher variables. Building upon this idea, [8] and [9] apply the Burkholder method to online learning. By viewing the Rademacher bounds as a martingale inequality, the authors "port over" the Burkholder method from the theory of martingales into the theory of online learning, allowing existing research into martingale inequalities to inform the development of online learning algorithms.

There are quite a few steps involved in analyzing a learning problem via sequential Rademacher complexities or the Burkholder method. Furthermore, because these equivalences are indirect, even when the Rademacher bounds end up being tight (as under the conditions of Theorem 7 of [23] or Proposition 18 of [9]) the random variable itself is often a tricky object to analyze. It typically has a complicated dependence on the regret concept, and sometimes also requires considering the learner's strategy.

We can see how this indirect bound plays out in the simple setting of online linear optimization.

1.4.1 OLO Over the Euclidean Unit Ball

We first translate the bounded OLO setting to our notation: $\mathcal{Y} = \mathcal{B}$, $\mathcal{A} = \mathcal{B}$, $\ell(a, y) = -\langle a, y\rangle$, where $\mathcal{B}$ is the d-dimensional Euclidean unit ball. We consider ϕ corresponding to standard external regret $\phi(\mathbf{y}) = -\min_{a\in\mathcal{B}} \sum_{t=1}^{T} \ell(a, \mathbf{y}_t) = \left\|\sum_{t=1}^{T} \mathbf{y}_t\right\|$. [2] found that the minimax optimal regret for this setting (with $d \geq 3$) over T timesteps is $\sqrt{T}$. Using sequential Rademacher complexity, [19] (Chapter 10, Example 3) show that this regret is determined, up to a factor of 2, by the standard martingale inequality $\mathbb{E}\left\|\sum_{t=1}^{T} M_t\right\| \leq \sqrt{T}$ applied to the set of dyadic martingales. Furthermore, Theorem 7 of [23] allows us to skip the sequential Rademacher complexity argument and eliminate the factor of 2, but only because we are considering standard external regret. The analysis of adaptive regrets, as in [7], still requires Rademacher expectations.

We will revisit this setting in detail later. We will show that the tightness of the martingale

inequality *implies* the tightness of the learning guarantee. That is, we show an exact equality between stochastic expectations and the minimax value of learning games, unlike the inequality of [23]. We will also show tight martingale equivalences for a broad class of data-dependent ϕ. Specifically, we will analyze ϕ representing regret bounds which adapt to the norms of the inputs, showing a direct equivalence between these adaptive bounds and a martingale moment inequality, and thereby proving that a constant from [9] cannot be improved.

1.5 Contributions

[20] note that the direct equivalences found in Cover, and the indirect equivalences derived via Rademacher bounds, are qualitatively similar. In both cases, it is possible to say something meaningful about the minimax value of a learning game, by taking an expectation over a stochastic process which is stationary (either i.i.d. or a martingale difference sequence). This thesis deepens this connection. We will identify general conditions under which learning problems enjoy direct, Cover-like, stochastic-adversarial equivalences. In our equivalences, the relevant bounds are given by the worst-case expectation over, simply, $\phi(Y_{1:T})$, where $Y_{1:T}$ is an underlying stationary (i.i.d. or zero-mean) process governing Nature's outcomes.

As mentioned in the previous section, this is applicable to some commonly-studied settings, yielding some incremental novel results. We will extend Cover's results in discrete prediction settings, and also show that some continuous settings enjoy direct equivalences. We will extend the result of [2] in online linear optimization constrained to the unit ball under standard external regret to $d = 2$ (the 2-dimensional ball). We will also show that AdaGrad-style regret bounds (as in equation (6) of [5]), require a constant of at least 2 in front of the adaptive comparator term.

We also make a few more minor contributions along the way, including showing that a stability condition we analyze also characterizes implicit or "natural" constraints on the learner in OLO problems, suggesting (but not analyzing) some stochastic-adversarial equivalences for less-common comparator terms, and deriving a novel (but generally suboptimal) class of regret bounds in unconstrained online linear optimization.

Chapter 2

General Equivalence-Preserving Restrictions

We return to our motivating question from Sec. 1.1: When are adversarial regret bounds equal to the worst-case regret bounds for a Bayes-optimal learner over a restricted family of data-generating processes?

2.1 Setup

We assume that $\mathcal{Y}$ is a compact, convex, Polish space and that $\mathcal{A}$ is a convex set.[1] We also assume that $\ell(a, y) : \mathcal{A} \times \mathcal{Y} \to \mathbb{R}$ is bounded, that $\ell(\cdot, y)$ is convex for any fixed y, and that $\ell(a, \cdot)$ is Borel-measurable and upper semicontinuous for any fixed a. Throughout, $\Delta(X)$ refers to the probability simplex over a topological space X under the Borel measure.

2.1.1 Conditions on ℓ, $\mathcal{P}$, and ϕ

It is obvious that if the supremum over $\Delta(\mathcal{Y}^T)$ in eqn. (1.1) can be achieved or approached within $\mathcal{P}_{1:T}$, then we preserve our desired equivalence even when Nature is restricted to processes within $\mathcal{P}_{1:T}$. Our first result (and our main result for the general case) shows that we don't need to analyze the behavior of the whole process at every timestep. For certain kinds of stationary distributions, a condition on the tail behavior of the process suffices.

We first define what we mean by stationary:

[1] We could alternatively require $\mathcal{A}$ to be compact instead of $\mathcal{Y}$. There will be some discussion of this alternative in Sections 2.2.1 and 2.3.

Definition 1 ($\mathcal{P}$-stationary) Given a family of one-round distributions $\mathcal{P} \subseteq \Delta(\mathcal{Y})$, define the $\mathcal{P}$-stationary family of joint distributions $\mathcal{P}_{1:T} \subseteq \Delta(\mathcal{Y}^T)$ to be the family whose conditional distributions at each timestep are $\mathcal{P}$: $\mathcal{P}_{1:T} = \{P_{1:T} : (P_t \mid \mathbf{y}_{1:t-1}) \in \mathcal{P} \quad \forall t, \mathbf{y}_{1:t-1}\}$.

An important feature of $\mathcal{P}$-stationarity is that, regardless of the history $\mathbf{y}_{1:t-1}$ of the process, the allowable distributions $\mathcal{P}_{t:T}$ over the tails remain the same.

We will also constrain our family to ensure that the best response of a Bayes-optimal learner gives constant loss:

Definition 2 (ℓ-constant) A family of one-round distributions $\mathcal{P} \subseteq \Delta(\mathcal{Y})$ is *ℓ-constant* with respect to a loss function ℓ if the Bayes risk $L(P) = \inf_{a \in \mathcal{A}} \mathbb{E}_P [\ell(a, Y)]$ is identical for all $P \in \mathcal{P}$. We'll label this constant Bayes risk $L(\mathcal{P})$.

ℓ-constancy may or may not coincide with natural assumptions in statistical learning. In the bounded OLO setting, any $\mathcal{P}$ consisting of zero-mean distributions is ℓ-constant, an assumption corresponding, for example, to the arbitrage-free assumption in finance. In other settings, we might take $\mathcal{P}$ to be a singleton containing a worst-case distribution with maximal Bayes risk. However, we need not give $\mathcal{P}$ a statistical learning interpretation at all. We can just use it as a tool for analyzing adversarial problems, specifying $\mathcal{P}$ to line up with known results from stochastic processes.

For our desired equivalences to hold, we also need to place a somewhat complicated condition on the comparator function ϕ. We will express this condition with the help of the following two definitions:

Definition 3 (Worst-Case Tail Expectation) Given $T \in \mathbb{N}$, a measurable bounded-above function $\phi : \mathcal{Y}^T \to \mathbb{R}$, and the $\mathcal{P}$-stationary process $\mathcal{P}_{1:T} \subseteq \Delta(\mathcal{Y}^T)$, define

$$\hat{\phi}_t(\mathbf{y}_{1:t-1}, y) = \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T})] \tag{2.1}$$

In other words, $\hat{\phi}$ evaluates the expected value of ϕ, having fixed a given prefix, under the worst-case process in $\mathcal{P}_{1:T}$. Recall that we can drop the conditioning on $\mathbf{y}_{1:t-1}, y$ because of $\mathcal{P}$-stationarity.

Our main requirement on $\hat{\phi}$ aims to reduce the global optimality condition (that the supremum in eqn. (1.1) is achieved or approached in $\mathcal{P}_{1:T}$) to a per-round condition:

Definition 4 (Stable Function) A measurable, bounded-above function $f : \mathcal{Y} \to \mathbb{R}$ is *stable* with respect to ℓ on $\mathcal{P}$ if:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{2.2}$$

Note that if we have ℓ-constancy, we can write the right-hand-side as $L(\mathcal{P}) + \sup_{P \in \mathcal{P}} \mathbb{E}_P\left[f(Y)\right]$. Later on, in Section 2.3, we will interpret the stability condition in convex analysis terms.

2.2 Main Theorem

We are finally able to give our main result in the general setting.

Theorem 5 We are given $T \in \mathbb{N}$, a measurable, bounded function $\phi : \mathcal{Y}^T \to \mathbb{R}$ which is upper semicontinuous in each coordinate, an ℓ-constant family $\mathcal{P} \subseteq \Delta(\mathcal{Y})$ of conditional distributions, and the $\mathcal{P}$-stationary family $\mathcal{P}_{1:T} \subseteq \Delta(\mathcal{Y}^T)$ of joint distributions.

If for all $t \leq T$ and $\mathbf{y}_{1:t-1} \in \mathcal{Y}^{t-1}$, $f(y) = \hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is stable with respect to ℓ on $\mathcal{P}$, then

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = TL(\mathcal{P}) + \sup_{P_{1:T} \in \mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}}[\phi(Y_{1:T})] \tag{2.3}$$

where Alg_T is the set of all algorithms in $\mathcal{Y}^{<T} \to \mathcal{A}$.

The argument proceeds by a backwards induction. We use Ky Fan's minimax theorem (Theorem 2 in [6]) to swap the inf and the sup on each round, then use the stability condition to go from $\sup_{\Delta(\mathcal{Y})}$ to $\sup_{\mathcal{P}}$ at each round. Then, because we need to consider the entire tail behavior of ϕ at each round, we use ℓ-constancy and $\mathcal{P}$-stationarity to show that this tail remains stable, allowing the induction to continue. The full proof is in Appendix A.

The left-hand-side of eqn. (2.3) is exactly the value of our learning game. So, Theorem 5 allows us to exactly characterize the value of the game via a supremum over expectations. When our

conditions are met, we can use tools of probability theory to establish either an exact characteriation of the value of the game (if existing probability theory gives us an equality), or a bound on the value of the game (if existing probability theory gives us an inequality). And once we know the value of the game (or an upper bound on the value of the game), we can often use that to synthesize practical algorithms.[2] We won't synthesize any novel algorithms in this thesis, but we will use this direct equivalence to analyze novel binary prediction settings, as well as to show the sharpness of existing inequalities for the value of well-known online learning games.

The converse is true as well. The proof of a regret bound for a particular learning algorithm or learning problem, is *also* a proof of a stochastic inequality which must hold over the entire class $\mathcal{P}$. We will give some examples of such inequalities later on, in Section 4.5. (How useful these inequalities actually are, is up to the reader to decide...).

2.2.1 On the Applicability of Ky Fan's Minimax Theorem

To justify the use of the minimax theorem (and to help explain the conditions we put on the problem setup in 2.1), we check the conditions required by the minimax theorem. The key minimax step requires the following swap:

$$\inf_{a\in\mathcal{A}} \sup_{P\in\Delta(\mathcal{Y})} \mathbb{E}_P\left[\ell(a,Y)+\phi(\mathbf{y}_{1:T-1},Y)\right] = \sup_{P\in\Delta(\mathcal{Y})} \inf_{a\in\mathcal{A}} \mathbb{E}_P\left[\ell(a,Y)+\phi(\mathbf{y}_{1:T-1},Y)\right]$$

for a bounded, measurable, upper semi-continuous ϕ. For Ky Fan's minimax theorem to apply, one of the decision spaces must be a compact Hausdorff space. We must further check that, as a function of the player in the compact space, the objective is upper/lower semicontinuous (depending on whether the player is maximizing/minimizing, respectively). Finally, the objective must be convex for the minimizing player and concave for the maximizing player.

We check each of these conditions in turn, first under the standard set of assumptions in 2.1, then under an alternate set of assumptions.

[2] See [24], which establishes a meta-algorithm for going from game values to learning algorithms.

2.2.1.1 Primary Assumptions: $\mathcal{Y}$ Compact

Suppose, as we originally did in the setup in this chapter, that $\mathcal{Y}$ is a compact, convex, Polish space. Then, so too is $\Delta(\mathcal{Y}) \subset ca(X)$, where $ca(X)$ is the topological vector space of signed Borel measures of bounded variation, under the weak-* topology. See Section 14.1 of [3] for these results.

For conditions on the objective, first fix $a \in \mathcal{A}$. Then, the objective in $\Delta(\mathcal{Y})$ is:

$$f(P) : \Delta(\mathcal{Y}) \to \mathbb{R} = \mathbb{E}_P \left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y)\right]$$

f is linear, and therefore concave, by the linearity of expectations/integration. Because both $\ell(a, \cdot)$ and ϕ are bounded and upper semicontinuous, so is their sum and, by Theorem 14.5 of [3], so is the expected value functional.

Next fix $P \in \Delta(\mathcal{Y})$. The objective is:

$$f(a) : \mathcal{A} \to \mathbb{R} = \mathbb{E}_P \left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y)\right] = \mathbb{E}_P \left[\ell(a, Y)\right] + \mathbb{E}_P \left[\phi(\mathbf{y}_{1:T-1}, Y)\right]$$

The second term is a constant and doesn't matter for Ky Fan's theorem. f is convex by the convexity of ℓ and the property that a convex combination of bounded convex functions remains convex.

2.2.1.2 Secondary Assumptions: $\mathcal{A}$ Compact

Now we explore a different set of assumptions. Under these assumptions, we will be able to relax the requirement that ϕ is upper semicontinuous.

Specifically, suppose $\mathcal{A}$ is a compact, convex, first-countable, Hausdorff space, while $\mathcal{Y}$ is now just a convex (but not necessarily compact) Polish space. Instead of requiring $\ell(a, \cdot)$ to be upper semicontinuous, we require $\ell(\cdot, y)$ to be lower semicontinuous for all y.

For the objective, first fix $a \in \mathcal{A}$. Then, the objective is:

$$f(P) : \Delta(\mathcal{Y}) \to \mathbb{R} = \mathbb{E}_P \left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y)\right]$$

As before, f is linear, and therefore concave, by the linearity of expectations/integration.

Next fix $P \in \Delta(\mathcal{Y})$. As before, the objective is:

$$f(a) : \mathcal{A} \to \mathbb{R} = \mathbb{E}_P\left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y)\right] = \mathbb{E}_P\left[\ell(a, Y)\right] + \mathbb{E}_P\left[\phi(\mathbf{y}_{1:T-1}, Y)\right]$$

And again, the second term is a constant and doesn't matter for Ky Fan's theorem, and f is convex. Because ℓ is bounded, Fatou's Lemma guarantees that $\mathbb{E}_P\left[\ell(a, Y)\right]$ will be lower semicontinuous if $\ell(\cdot, y)$ is as well, for all y:

$$\begin{aligned}
& \ell \text{ is l.s.c.} \\
\implies \quad & \forall a, \{a_n\}_{n=0}^{\infty} : \lim_{n\to\infty} a_n = a \quad \ell(a, y) \leq \liminf_{n\to\infty} \ell(a_n, y) \\
\implies \quad & \forall a, \{a_n\}_{n=0}^{\infty} : \lim_{n\to\infty} a_n = a \quad \mathbb{E}_P\left[\ell(a, Y)\right] \leq \mathbb{E}_P\left[\liminf_{n\to\infty} \ell(a_n, Y)\right]
\end{aligned}$$

By Fatou's,

$$\implies \quad \forall a, \{a_n\}_{n=0}^{\infty} : \lim_{n\to\infty} a_n = a \quad \mathbb{E}_P\left[\ell(a, Y)\right] \leq \liminf_{n\to\infty} \mathbb{E}_P\left[\ell(a_n, Y)\right]$$

Thus, we can apply Ky Fan's minimax theorem under this alternate set of assumptions as well, and recover Theorem 5.

2.3 Convex Analysis Characterization of Stable Functions

In the remainder of this chapter, we present general results about stable functions.

Our general result ends up having a somewhat awkward presentation to accommodate the full generality of the problem. Perhaps the key takeaway from this section is that under some additional problem constraints (explored in Proposition 7), stability is equivalent to the condition (2.6):

$$\exists P^* \in \mathcal{P} : \quad f \in \partial(-L)(P^*) + N_{\Delta(\mathcal{Y})}(P^*)$$

Where ∂ is the subdifferential, and $N_{\Delta(\mathcal{Y})}$ is the normal cone to the simplex $\Delta(\mathcal{Y})$ at P^*. That is, we can characterize stable functions in terms of the superdifferential of the Bayes risk (subdifferential of the negative Bayes risk) at some particular distribution in $\mathcal{P}$, adjusted by some element in the normal cone. This gives us an algorithm for finding all stable functions: we choose a particular P^*, then choose a particular subgradient of $-L$ at P^*, add a constant, then subtract off an arbitrary nonnegative function over the parts of the domain which have no support on P^*.

The fully general characterization requires a generalization of the subdifferential to an ϵ-subdifferential ∂_ϵ:

$$g \in \partial_\epsilon(f)(x) \iff f(y) \geq f(x) + \langle g, y - x \rangle - \epsilon \quad \forall y \tag{2.4}$$

See [25], Section 23, for a complete treatment of the ϵ-subdifferential. Comparing to the definition of the standard subdifferential:

$$g \in \partial(f)(x) \iff f(y) \geq f(x) + \langle g, y - x \rangle \quad \forall y$$

we see that the ϵ-subdifferential is a slightly relaxed definition which allows for being "close" to a perfect subgradient.

This allows for us to deal with $\mathcal{P}$ which aren't closed, for example, which are missing a single point in the domain. See Fig. 2.1

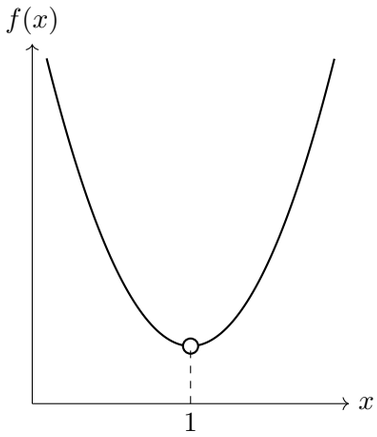


Figure 2.1: Why the ϵ-subdifferential is useful: Suppose $\mathcal{P}$ is parameterized by a single real x. If $x = 1$ is missing from $\mathcal{P}$, we will be unable to find an x^* such that $0 \in \partial(f)(x^*)$. However, for any ϵ, we can find an x^* close to 1 such that $0 \in \partial_\epsilon(f)(x^*)$.

With that background, we can present our general characterization of stability in terms of ϵ-subdifferentials:

Proposition 6 A measurable bounded function $f : \mathcal{Y} \to \mathbb{R}$ is stable with respect to ℓ on $\mathcal{P}$ if and only if, for all $\epsilon > 0$, there exist $P^* \in \mathcal{P}$ and $\epsilon_L \in (0, \epsilon)$ such that:

$$f \in \partial_{\epsilon_L}(-L)(P^*) + \partial_{(\epsilon-\epsilon_L)}\delta_{\Delta(\mathcal{Y})}(P^*) \tag{2.5}$$

where $+$ is the Minkowski sum, L is the Bayes risk, ∂_ϵ is the ϵ-subdifferential, and δ is the standard $0/\infty$ indicator function of convex analysis, so that $\partial_{\epsilon-\epsilon_L}\delta_{\Delta(\mathcal{Y})}$ is the $(\epsilon - \epsilon_L)$-normal cone to the simplex.

Proof We turn the stability condition (2.2) into an unconstrained optimality condition:

$$\sup_{P\in\Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P\in\mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right)$$

$$\iff \quad \sup_{P\in\mathcal{M}(\mathcal{Y})} \left(L(P) + \langle P, f\rangle - \delta_{\Delta(\mathcal{Y})}(P)\right) \leq \sup_{P\in\mathcal{P}} \left(L(P) + \langle P, f\rangle - \delta_{\Delta(\mathcal{Y})}(P)\right)$$

where $\mathcal{M}(\mathcal{Y})$ is the space of finite signed Borel measures over $\mathcal{Y}$, and $\langle\cdot,\cdot\rangle$ is the standard dual pairing defined by integration: $\langle P, f\rangle = \int_{\mathcal{Y}} f dP$. Note we dropped the trivial direction of inequality implied by $\mathcal{P} \subseteq \mathcal{M}(\mathcal{Y})$. We express this relationship in terms of a pointwise inequality. Stability is equivalent to the claim that for all $\epsilon > 0$, there exists $P^* \in \mathcal{P}$ such that, for all $P \in \mathcal{M}(\mathcal{Y})$:

$$L(P) + \langle P, f\rangle - \delta_{\Delta(\mathcal{Y})}(P) \leq L(P^*) + \langle P^*, f\rangle - \delta_{\Delta(\mathcal{Y})}(P^*) + \epsilon$$

$$(-L)(P^*) + \delta_{\Delta(\mathcal{Y})}(P^*) - \epsilon + \langle P - P^*, f\rangle \leq (-L)(P) + \delta_{\Delta(\mathcal{Y})}(P)$$

$$(-L + \delta_{\Delta(\mathcal{Y})})(P^*) - \epsilon + \langle P - P^*, f\rangle \leq (-L + \delta_{\Delta(\mathcal{Y})})(P)$$

This is the definition of ϵ-subdifferential, as in eqn. (2.4) We can apply the addition calculus for ϵ-subdifferentials from [11] (noting that the intersection of the relative interiors of the domains of $-L$ and $\delta_{\Delta(\mathcal{Y})}$ is non-empty) to yield our final result. ■

If we impose our main assumptions, that $\mathcal{Y}$ is compact, $\ell(a, \cdot)$ is upper semi-continuous, and f is upper semi-continuous, and additionally assume that $\mathcal{P}$ is closed, we can somewhat simplify the presentation of the above proposition:

Proposition 7 Assume $\mathcal{Y}$ is compact, and $\mathcal{P}$ is sequentially closed under the weak-* topology on $\Delta(\mathcal{Y})$. That is, for every sequence of distributions $\{P_n\}_{n=1}^{\infty} \subseteq \mathcal{P}$ satisfying the convergence property:

$$\lim_{n\to\infty} \mathbb{E}_{P_n}[f(Y)] = \mathbb{E}_P[f(Y)] \quad \forall f \in C_b(\mathcal{Y})$$

where $C_b(\mathcal{Y})$ is the set of bounded continuous functions $\mathcal{Y} \to \mathbb{R}$, assume that $P \in \mathcal{P}$. Then, a measurable, bounded, upper semicontinuous (u.s.c.) function $f : \mathcal{Y} \to \mathbb{R}$ is stable with respect to ℓ on $\mathcal{P}$ if and only if there exists $P^* \in \mathcal{P}$ such that:

$$f \in \partial(-L)(P^*) + N_{\Delta(\mathcal{Y})}(P^*) \tag{2.6}$$

where $+$ is the Minkowski sum, L is the Bayes risk, and $N_{\Delta(\mathcal{Y})}(P^*)$ is the normal cone to the simplex at P^*.

The proof is somewhat verbose and can be found in Appendix A.

Although we won't directly use these results to prove the stability of subsequent ϕ, we will use it to motivate our choice of $\mathcal{P}$ in different settings. The most interesting stochastic-adversarial equivalences arise when ℓ and $\mathcal{P}$ coincide such that:

(1) $\mathcal{P}$ is ℓ-constant,

(2) $\mathcal{P}$ has enough structure to admit meaningful in-expectation bounds, and

(3) $\cap_\epsilon \partial_\epsilon(-L)(\mathcal{P})$ is a large set (e.g. if L is not differentiable somewhere in $\mathcal{P}$).

Note that the first two concerns tend to restrict $\mathcal{P}$ to smaller sets, so we can usually expect $\mathcal{P}$ to exclusively contain kinks (or approximate kinks) of L.

2.4 Stable Functions With Respect to i.i.d. $\mathcal{P}$

In stochastic-adversarial equivalences where the stochastic assumption is i.i.d., $\mathcal{P} = \{P_0\}$ is a singleton. In this case, we can use a simpler sufficient condition for stability:

Proposition 8 (Under Singleton $\mathcal{P}$, Per-Round Stability Suffices for Stability of $\hat{\phi}$) If, for all outcomes $\mathbf{y}_{-t} \in \mathcal{Y}^{T-1}$ and rounds $t \leq T$, $\phi_{\mathbf{y}_{-t}}(y) = \phi(\mathbf{y}_{1:t-1}, y, \mathbf{y}_{t+1:T})$ is stable with respect to ℓ and $\mathcal{P} = \{P_0\}$, then $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is also stable for all prefixes $\mathbf{y}_{1:t-1} \in \mathcal{Y}^{t-1}$.

Proof Suprema over $\mathcal{P}$ can be replaced by just P_0, so we have:

$$\hat{\phi}_t(\mathbf{y}_{1:t-1}, y) = \mathbb{E}_{P_0^{T-t}}[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T})]$$

and the stability condition for $\phi_{\mathbf{y}_{-t}}$ becomes:

$$\sup_{P \in \Delta\mathcal{Y}} (L(P) + \mathbb{E}_P[\phi(\mathbf{y}_{1:t-1}, Y, \mathbf{y}_{t+1:T})]) = L(P_0) + \mathbb{E}_{P_0}[\phi(\mathbf{y}_{1:t-1}, Y, \mathbf{y}_{t+1:T})]$$

We'll break down the global objective for $\hat{\phi}_t$ given a prefix $\mathbf{y}_{1:t-1}$:

$$\begin{aligned}
\sup_{P \in \Delta\mathcal{Y}} \left(L(P) + \mathbb{E}_P \left[\hat{\phi}_t(\mathbf{y}_{1:t-1}, Y) \right] \right) &= \sup_{P \in \Delta\mathcal{Y}} \left(L(P) + \mathbb{E}_P \left[\mathbb{E}_{P_0^{T-t}} \left[\phi(\mathbf{y}_{1:t-1}, Y, Y_{t+1:T}) \right] \right] \right) \\
&= \sup_{P \in \Delta\mathcal{Y}} \left(L(P) + \mathbb{E}_{P_0^{T-t}} \left[\mathbb{E}_P \left[\phi(\mathbf{y}_{1:t-1}, Y, Y_{t+1:T}) \right] \right] \right)
\end{aligned}$$

By Jensen's inequality:

$$\leq \mathbb{E}_{P_0^{T-t}} \left[\sup_{P \in \Delta\mathcal{Y}} \left(L(P) + \mathbb{E}_P \left[\phi(\mathbf{y}_{1:t-1}, Y, Y_{t+1:T}) \right] \right) \right]$$

Applying the per-round stability condition:

$$
\begin{aligned}
&= \mathbb{E}_{P_0^{T-t}}\left[L(P_0) + \mathbb{E}_{P_0}[\phi(\mathbf{y}_{1:t-1}, Y, Y_{t+1:T})]\right] \\
&= L(P_0) + \mathbb{E}_{P_0}\left[\mathbb{E}_{P_0^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, Y, Y_{t+1:T})\right]\right] \\
&= L(P_0) + \mathbb{E}_{P_0}\left[\hat{\phi}_t(\mathbf{y}_{1:t-1}, Y)\right] \\
&= \sup_{P \in \mathcal{P}}\left(L(P) + \mathbb{E}_{P}\left[\hat{\phi}_t(\mathbf{y}_{1:t-1}, Y)\right]\right)
\end{aligned}
$$

The other direction of the inequality is trivial, so we have our optimality condition for $\hat{\phi}_t$. ■

We'll use this result to simplify some results in the binary prediction setting in Chapter 3.

2.4.1 Under Singleton $\mathcal{P}$, Per-Round Stability Is Not Necessary for Stability of $\hat{\phi}$

The condition of Proposition 8 is sufficient, but it is not necessary, as the following example demonstrates: (We will cheat slightly and use a result from later in the thesis, but this does not introduce any circularity).

Example 1 Let $\mathcal{Y} = \{0, 1\}$, $\mathcal{P} = \{\text{Ber}(1/2)\}$, $T = 2$, with $\mathcal{A} = [0, 1]$, evaluated under 0-1 loss. Take:

$$
\phi(y_1, y_2) = \begin{cases} 0 & y_1 = 0, y_2 = 0 \\ 1 & y_1 = 0, y_2 = 1 \\ 2 & y_1 = 1, y_2 = 0 \\ 1 & y_1 = 1, y_2 = 1 \end{cases}
$$

Then $\phi(\cdot, 0)$ is not stable, but $\hat{\phi}$ is stable for all prefixes.

Proof We use a result from later (found within the proof of Proposition 10), namely that the stability condition for a function $f(y)$ in this setting is given by:

$$
|f(1) - f(0)| \leq 1
$$

We won't prove this yet, but even now, it should strike us as highly plausible when we recall Cover's stability condition in the same setting (see eqn. (1.2)). From this, we can immediately see that $\phi(\cdot, 0)$ is not stable, since $2 - 0 > 1$. Then, the stability of $\hat{\phi}$ in this setting reduces to:

$$\left|\mathbb{E}_{\text{Ber}(1/2)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 1, Y_{t+1:T})\right] - \mathbb{E}_{\text{Ber}(1/2)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 0, Y_{t+1:T})\right]\right| \leq 1$$

for any $t \in \{1, 2\}$ and any prefix $\mathbf{y}_{1:t-1}$. Splitting this up case-by-case gives us:

$$\begin{aligned} |\phi(1,0) + \phi(1,1) - \phi(0,0) - \phi(0,1)| &\leq 2 \\ |\phi(0,1) - \phi(0,0)| &\leq 1 \\ |\phi(1,1) - \phi(1,0)| &\leq 1 \end{aligned}$$

We can easily verify that these conditions are all satisfied by our chosen ϕ. ■

2.5 Per-Round Stability Is Not Generally Sufficient for Stability of $\hat{\phi}$

The simplification of the stability condition in Section 2.4 does not, unfortunately, hold when we allow $\mathcal{P}$ to contain more than one element. $\phi(\mathbf{y}_{1:t-1}, y, \mathbf{y}_{t+1:T})$ can be stable for all fixed outputs $\mathbf{y}_{-t}$, while $\hat{\phi}_t(\mathbf{y}_{1:t-1})$ is not. Essentially, because $\hat{\phi}_t(\mathbf{y}_{1:t_1}, y)$ is a worst-case expectation over tail distributions in $\mathcal{P}_{t+1:T}$, changing the input y can alter which distribution in $\mathcal{P}_{t+1:T}$ is the worst-case. We illustrate this with two examples, which may also help us develop an understanding of the stability condition.

Example 2 Let $\mathcal{Y} = \{-1, 1\}$, $\mathcal{P} = \{\delta_{-1}, \delta_1\}$ (where δ_y is the Dirac delta distribution at y), $L(P) = -(\mathbb{E}_P[Y])^2$, $T = 2$, and $\phi = 2y_1y_2$. Then $\phi(\mathbf{y}_1, \cdot)$ and $\phi(\cdot, \mathbf{y}_2)$, for any $\mathbf{y} \in \mathcal{Y}$, are stable with respect to ℓ and $\mathcal{P}$. However, $\hat{\phi}_1(\cdot)$ is not stable.

Proof We begin by showing the stability of ϕ. Define $f_1(y) = \phi(1, y) = \phi(y, 1) = 2y$, so:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f_1(Y)\right]\right) = \sup_{P \in \Delta(\mathcal{Y})} \left(-(\mathbb{E}_P[Y])^2 + \mathbb{E}_P\left[f_1(Y)\right]\right)$$

Representing each possible distribution over $\mathcal{Y}$ as a Bernoulli distribution which selects $Y = 1$ with probability p:

$$= \sup_{p \in [0,1]} \left(-(2p-1)^2 + 2(2p-1)\right)$$

The quadratic is maximized at $2p - 1 = 1$, so

$$= -(\mathbb{E}_{\delta_1}[Y])^2 + \mathbb{E}_{\delta_1}\left[f_1(Y)\right] \leq \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f_1(Y)\right]\right)$$

Similarly, define $f_{-1}(y) = \phi(-1, y) = \phi(y, -1) = -2y$, so:

$$\begin{aligned}\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f_{-1}(Y)\right]\right) &= \sup_{P \in \Delta(\mathcal{Y})} \left(-(\mathbb{E}_P[Y])^2 + \mathbb{E}_P\left[f_{-1}(Y)\right]\right) \\ &= \sup_{p \in [0,1]} \left(-(2p-1)^2 - 2(2p-1)\right)\end{aligned}$$

This quadratic is maximized at $2p - 1 = -1$, or $p = 0$, so

$$= -(\mathbb{E}_{\delta_{-1}}[Y])^2 + \mathbb{E}_{\delta_{-1}}\left[f_{-1}(Y)\right] \leq \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f_{-1}(Y)\right]\right)$$

Thus, f_1 and f_{-1} are both stable, so ϕ is also stable with respect to any fixed argument in either position.

However,

$$\hat{\phi}_1(y_1) = \sup_{P \in \{\delta_1, \delta_{-1}\}} \mathbb{E}_P[\phi(y_1, Y_2)] = \max(2y_1, -2y_1) = 2|y_1| = 2$$

So,

$$\sup_{P\in\Delta(\mathcal{Y})}\left(L(P)+\mathbb{E}_P\left[\hat{\phi}_1(Y)\right]\right)=\sup_{P\in\Delta(\mathcal{Y})}\left(-(\mathbb{E}_P[Y])^2+\mathbb{E}_P\left[\hat{\phi}_1(Y)\right]\right)$$
$$=\sup_{p\in[0,1]}\left(-(2p-1)^2+2\right)=2$$

But,

$$\sup_{P\in\mathcal{P}}\left(L(P)+\mathbb{E}_P\left[\hat{\phi}_1(Y)\right]\right)=\sup_{P\in\mathcal{P}}\left(-(\mathbb{E}_P[Y])^2+\mathbb{E}_P\left[\hat{\phi}_1(Y)\right]\right)$$
$$=\sup_{y\in\{-1,1\}}\left(-y^2+2\right)=1$$

So $\hat{\phi}_1$ is not stable. ■

One might remain skeptical. Perhaps the non-convexity of $\mathcal{P}$ in the last example might be behind its failure. However, we can construct somewhat more involved examples, in which $\mathcal{P}$ is convex.

Example 3 Let $\mathcal{Y}=\{1,2,3\}$, $\mathcal{P}=\{(p_1,p_2,1-(p_1+p_2)):p_1\leq 0.2,p_2\leq 0.2\}$, $L(P)=\min(0,2-10|p_1-p_2|)$, $T=2$, and:

$$\phi(y_1,y_2)=\begin{cases}0 & y_1=3 \text{ or } y_2=3\\ 10 & \text{o.w. and } y_1=y_2\\ -10 & \text{o.w.}\end{cases}$$

Then $\phi(\mathbf{y}_1,\cdot)$ and $\phi(\cdot,\mathbf{y}_2)$ are stable with respect to ℓ and $\mathcal{P}$ (and for any choice of $\mathbf{y}_1$ or $\mathbf{y}_2$), but $\hat{\phi}_1$ is not stable.

Proof Again, we begin by showing the stability of ϕ. Define

$$f_1(y) = \phi(1, y) = \phi(y, 1) = \begin{cases} 10 & y = 1 \\ -10 & y = 2 \\ 0 & y = 3 \end{cases}$$

so that

$$\begin{aligned}
\sup_{P \in \Delta(\mathcal{Y})} (L(P) + \mathbb{E}_P [f_1(Y)]) &= \sup_{P \in \Delta(\mathcal{Y})} (\min(0, 2 - 10|p_1 - p_2|) + \mathbb{E}_P [f_1(Y)]) \\
&= \sup_{p_1, p_2} (\min(0, 2 - 10|p_1 - p_2|) + 10(p_1 - p_2)) \\
&= \sup_{p_1, p_2} \left(\begin{cases} \min(10(p_1 - p_2), 2) & p_1 \geq p_2 \\ \min(10(p_1 - p_2), 20(p_1 - p_2)) & \text{o.w.} \end{cases} \right) \\
&= \sup_{p_1, p_2} \left(\begin{cases} \min(10(p_1 - p_2), 2) & p_1 \geq p_2 \\ 20(p_1 - p_2) & \text{o.w.} \end{cases} \right) \\
&\leq 2
\end{aligned}$$

This upper bound is realized in $\mathcal{P}$ by $p_1 = 0.2$ and $p_2 = 0$, so $f_1(y)$ is stable. The argument for $f_2(y)$ is symmetric, with p_1 and p_2 flipped. Finally, define

$$f_3(y) = \phi(3, y) = \phi(y, 3) = 0$$

$$\begin{aligned}
\sup_{P \in \Delta(\mathcal{Y})} (L(P) + \mathbb{E}_P [f_3(Y)]) &= \sup_{P \in \Delta(\mathcal{Y})} (\min(0, 2 - 10|p_1 - p_2|)) \\
&\leq 0
\end{aligned}$$

This upper bound (of 0) is realized in $\mathcal{P}$ by $p_1 = 0.2$ and $p_2 = 0$, so f_3 is stable. Thus, f_1, f_2, and f_3

are all stable, so $\phi(\mathbf{y}_1, \cdot)$ and $\phi(\cdot, \mathbf{y}_2)$ are stable as well.

However,

$$\hat{\phi}_1(y_1) = \sup_{P \in \mathcal{P}} \mathbb{E}_P[\phi(y_1, Y_2)] = \begin{cases} \mathbb{E}_{(0.2,0,0.8)}\phi(1, Y) = 2 & y_1 = 1 \\ \mathbb{E}_{(0,0.2,0.8)}\phi(2, Y) = 2 & y_1 = 2 \\ 0 & y_1 = 3 \end{cases}$$

So,

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P \left[\hat{\phi}_1(Y) \right] \right) = \sup_{p_1, p_2} \min(0, 2 - 10|p_1 - p_2|) + 2(p_1 + p_2) \leq 2$$

But this upper bound of 2 is realized by $p_1 = p_2 = 0.5$, while any $p_1, p_2 \in \mathcal{P}$ will achieve at most 0.8, so $\hat{\phi}_1$ is not stable. ■

These discouraging counterexamples don't necessarily doom us to having to constantly analyze $\hat{\phi}$ instead of ϕ itself. Depending on the specific setting, there might be more general properties (e.g. Lipschitz continuity), which are preserved across the operation of taking worst-case tail expectations, and which are also sufficient for stability. In Section 4.2, we will identify such a general property for the setting of bounded online linear optimization.

Chapter 3

Binary Prediction

Recall from Section 1.2 that Thomas Cover, in [4], established a prototypical stochastic-adversarial equivalence in the binary prediction setting. In this chapter, we will explore how our more general Theorem 5 applies in the binary prediction setting. We'll recover Cover's result and extend it to different loss functions, illustrating the broad flexibility of the framework from Chapter 2.

Although the loss function ℓ and report space $\mathcal{A}$ will vary between the two learning problems we will consider, the outcome space for both is $\mathcal{Y} = \{0, 1\}$.

Let $\mathcal{P} = \{P_0\}$ be a singleton and $\mathcal{P}_{1:T} = P_0^T$. Note that this corresponds to a standard i.i.d. assumption. We get ℓ-constancy for free since $\mathcal{P}$ is a singleton. In the binary prediction setting, $\Delta(\mathcal{Y})$ can be parameterized as $\Delta(\mathcal{Y}) = \{\text{Ber}(p) : p \in [0, 1]\}$, where $\text{Ber}(p)$ is the Bernoulli distribution which selects 1 with probability p and 0 with probability $1 - p$.

We first find a straightforward simplification of the stability condition in this setting.

Lemma 9 Given some report space $\mathcal{A}$, a function $f : \{0, 1\} \to \mathbb{R}$ is stable with respect to $\ell : \mathcal{A} \times \{0, 1\} \to \mathbb{R}$ and $\mathcal{P} = \{\text{Ber}(p)\}$ if and only if:

$$\forall q \in [0, 1] \quad L(\text{Ber}(q)) - L(\text{Ber}(p)) \leq (p - q)\,(f(1) - f(0)) \tag{3.1}$$

Proof Begin with the stability condition (2.2):

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P \left[f(Y) \right] \right) = \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P \left[f(Y) \right] \right)$$

$$\sup_{q \in [0,1]} \left(L(\text{Ber}(q)) + \mathbb{E}_{\text{Ber}(q)} \left[f(Y) \right] \right) = L(\text{Ber}(p)) + \mathbb{E}_{\text{Ber}(p)} \left[f(Y) \right]$$

$$\sup_{q \in [0,1]} \left(L(\text{Ber}(q)) + qf(1) + (1-q)f(0) \right) = L(\text{Ber}(p)) + pf(1) + (1-p)f(0)$$

$$\forall q \in [0,1] \quad L(\text{Ber}(q)) - L(\text{Ber}(p)) \leq (p-q)f(1) + (q-p)f(0)$$

$$\forall q \in [0,1] \quad L(\text{Ber}(q)) - L(\text{Ber}(p)) \leq (p-q)\left(f(1) - f(0) \right)$$

■

We'll use this lemma to derive stability conditions for stochastic-adversarial equivalences under different distributions $\mathcal{P}$. Our two settings will be simple randomized prediction, in which the learner randomizes between predicting 0 and predicting 1 at each timestep, and an abstention setting, in which the learner can randomize between three predictions: zero, one, or "abstain."

3.1 Simple Random Prediction

In the simple setting, $\mathcal{A} = [0,1]$, representing the parameter for a Bernoulli distribution by which the learner makes a random prediction. We analyze a general cost-sensitive loss, in which false negatives and false positives may be penalized unequally according to some bias c. If the cost is balanced ($c = 1/2$), then this scenario is standard 0-1 loss, as studied by [4] and [21].

Proposition 10 (Stochastic-Adversarial Equivalence for Cost-Sensitive Loss) Under the cost-sensitive loss with bias $c \in [0,1]$:

$$\ell(a,y) = c \cdot a \mathbb{1}\{y = 0\} + (1-c)(1-a)\mathbb{1}\{y = 1\} = ca(1-y) + (1-c)(1-a)y \tag{3.2}$$

If, for all prefixes $\mathbf{y}_{1:t-1} \in \{0,1\}^{t-1}$:

$$\mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 1, Y_{t+1:T})\right] - \mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 0, Y_{t+1:T})\right] \in [c-1, c] \tag{3.3}$$

then we can express the exact minimax value of the prediction game over T steps:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T}) = Tc(1-c) + \mathbb{E}_{\mathrm{Ber}(c)^T}\left[\phi(Y_{1:T})\right] \tag{3.4}$$

In particular, if if condition (3.3) holds and $\mathbb{E}_{\mathrm{Ber}(c)^T}\left[-\phi(Y_{1:T})\right] = Tc(1-c)$, then $-\phi$ represents an achievable data-dependent loss for the learner.

Proof We find the Bayes risk:

$$L(\mathrm{Ber}(p)) = \min_{a\in[0,1]} \mathbb{E}_{\mathrm{Ber}(p)}\left[\ell(a, Y)\right] = \min_{a\in[0,1]} \left(c(a-p) + p(1-a)\right) = \min\left((1-c)p, c(1-p)\right)$$

Take $\mathcal{P} = \{\mathrm{Ber}(c)\}$, and note $L(\mathrm{Ber}(c)) = c(1-c)$. We want to apply Theorem 5 to the above ℓ and $\mathcal{P}$, by first showing that eqn. (3.3) is equivalent to our stability condition. The result will then follow immediately. So, our stability condition (from Lemma 9) is:

$$\begin{aligned}
\forall q \in [0,1] \quad & L(\mathrm{Ber}(q)) - L(\mathrm{Ber}(c)) \le (c-q)\left(f(1) - f(0)\right) \\
\forall q \in [0,1] \quad & \min((1-c)q, (1-q)c) - c(1-c) \le (c-q)\left(f(1) - f(0)\right) \\
\forall q \in [0,1] \quad & \min((1-c)(q-c), (c-q)c) \le (c-q)\left(f(1) - f(0)\right)
\end{aligned}$$

We split the cases $c \ge q$ and $c < q$. In both cases, we divide the inequality by $(c-q)$ (removing the dependence on q). This yields the two inequalities:

$$\min(c-1, c) \le f(1) - f(0) \qquad\qquad -\min(1-c, -c) \ge f(1) - f(0)$$

Combining gives us that stability is equivalent to: $f(1) - f(0) \in [c-1, c]$.

By definition, $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y) = \mathbb{E}_{\text{Ber}(c)^{T-t}}[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T})]$. Applying Theorem 5, we get:

$$\inf_{\text{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = TL(\text{Ber}(c)) + \mathbb{E}_{\text{Ber}(c)^T}[\phi(Y_{1:T})] \tag{3.5}$$

$$= Tc(1-c) + \mathbb{E}_{\text{Ber}(c)^T}[\phi(Y_{1:T})] \tag{3.6}$$

■

Note the high-level strategy used in the proof: identify the Bayes risk L for the given loss, choose a singleton $\mathcal{P} = \{P\}$ at a kink in L (in this case, $P = \text{Ber}(c)$), simplify the stability condition for the given L and $\mathcal{P}$, and apply Theorem 5. The principle of choosing $\mathcal{P}$ based on kinks in $L(P)$ is illustrated in Fig. 3.1.

Taking $c = 1/2$ and scaling by 2 recovers standard 0-1 loss. Under this formulation, Proposition 10 recovers one direction of result VI from [4].[1]

We can also apply Proposition 8, which immediately yields:

Corollary 11 Proposition 10 holds if we replace the tail stability condition (3.3) with the per-round condition:

$$\forall \mathbf{y}_{-t} \in \{0,1\}^{T-1} \quad \phi(\mathbf{y}_{1:t-1}, 1, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, 0, \mathbf{y}_{t+1:T}) \in [c-1, c] \tag{3.7}$$

This (cleaner but more restrictive) per-round stability condition is what [21] use in their extensions of Cover's result for 0-1 loss.

3.2 Abstain Loss

In this setting, $\mathcal{A} = [0,1] \times [0,1]$, the first element of which represents the probability the learner abstains from predicting, and the second element of which represents the probability the learner chooses 1, if they do decide to predict.

[1] We expect the opposite (only if) direction of that result should be achievable with just a little extra work, and make note of this as a possible extension in 6.2.2.

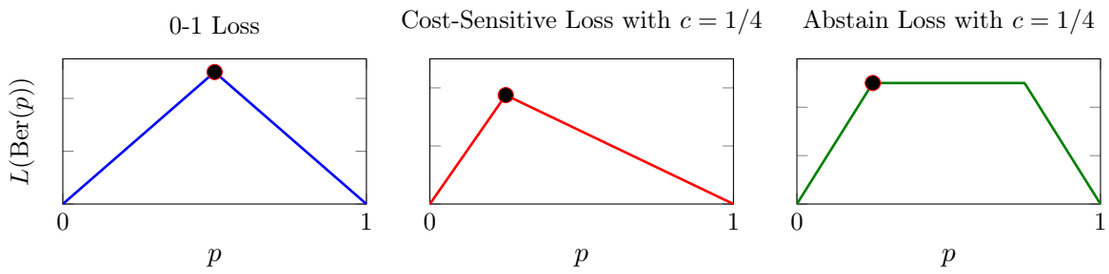


Figure 3.1: Per-round Bayes risk $L(\mathrm{Ber}(p))$ for three losses in the binary prediction setting (with abstention allowed in the third): 0-1 loss, cost-sensitive loss with $c = 1/4$, and abstain loss with cost of abstention $c = 1/4$. Highlighted are the kinks we chose for our singleton $\mathcal{P}$.

Proposition 12 (Stochastic-Adversarial Equivalence for Abstain Loss) Under the abstain loss, with the cost of abstention equal to $c \in [0, 1/2]$:

$$\ell((a, b), y) = ac + (1 - a)\left(b\mathbb{1}\{y = 0\} + (1 - b)\mathbb{1}\{y = 1\}\right) \tag{3.8}$$

If, for all prefixes $\mathbf{y}_{1:t-1} \in \{0, 1\}^{t-1}$:

$$\mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 1, Y_{t+1:T})\right] - \mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 0, Y_{t+1:T})\right] \in [-1, 0] \tag{3.9}$$

then we can express the exact minimax value of the prediction game over T steps:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T}) = Tc + \mathbb{E}_{\mathrm{Ber}(c)^T}\left[\phi(Y_{1:T})\right] \tag{3.10}$$

In particular, if condition (3.9) holds and $\mathbb{E}_{\mathrm{Ber}(c)^T}\left[-\phi(Y_{1:T})\right] = Tc$, then $-\phi$ represents an achievable data-dependent loss for the learner.

The proof (in Appendix B), follows the same strategy as Proposition 10: find the Bayes risk L, choose a single P at a kink in L (in this case, we chose $\mathrm{Ber}(c)$, although $\mathrm{Ber}(1 - c)$ would work as well), simplify the stability condition, and apply Theorem 5. And, again, we can apply Proposition 8, to get:

Corollary 13 Proposition 12 holds if we replace the tail stability condition (3.9) with the per-round condition:

$$\forall \mathbf{y}_{-t} \in \{0,1\}^{T-1} \quad \phi(\mathbf{y}_{1:t-1}, 1, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, 0, \mathbf{y}_{t+1:T}) \in [-1, 0] \tag{3.11}$$

Note that we chose $\mathcal{P} = \{\text{Ber}(c)\}$ somewhat arbitrarily, since $\text{Ber}(1-c)$ is also a non-differentiable point of the Bayes risk. Had we chosen $\text{Ber}(1-c)$, we would have found a stability condition of the form $f(1) - f(0) \in [0, 1]$. We could have even chosen $\mathcal{P} = \{\text{Ber}(c), \text{Ber}(1-c)\}$ to include both kinks. Had we done so, our stability condition would have been different (of the form $|\hat{\phi}_t(\mathbf{y}_{1:t-1}, 1) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, 0)| \leq 1$), but both the stability condition and the expression for minimax regret would have been much harder to understand. See Proposition 36 in Appendix B for details.

3.3 Multiclass Prediction

We don't thoroughly analyze the multiclass prediction setting, but expect that the results from [21] in this area should be recoverable via our framework. This result should additionally hold for an updated stability condition which requires only tail stability (that is, stability of $\hat{\phi}$), rather than per-round stability of ϕ.

Chapter 4

Online Linear Optimization Constrained to the Euclidean Ball

Section 1.3 briefly discussed the observation made by [12] that we can find direct equivalences between the value of unconstrained linear learning games and the expectation of the comparator term under outcomes distributed according to the worst-case martingale difference sequence.

Theorem 5 gives us tools to extend this observation to the constrained setting. By studying the stability condition in continuous linear optimization, we can find sufficient conditions for direct equivalences even when the learner is constrained.

Interestingly, as we will show in Section 4.7, the general condition for stability that we found implies that the constraints aren't "active" for an optimal learner. That is, if the comparator term ϕ is stable, then not only can we derive a stochastic-adversarial equivalence, but we can *also* guarantee that an optimal learner will make predictions inside the unit ball, whether or not they are explicitly constrained to do so by the problem setting. This insight could potentially be independently useful.

4.1 Setting

Take $\mathcal{Y} = \mathcal{A} = \mathcal{B}$, the Euclidean unit ball in d dimensions, and let $\ell(a, y) = -\langle a, y \rangle$.[1]

Throughout, we will make use of the equality:

$$L(P) = \inf_{a \in \mathcal{B}} \mathbb{E}_P \left[-\langle a, Y \rangle \right] = -\sup_{a \in \mathcal{B}} \langle a, \mathbb{E}_P \left[Y \right] \rangle = -\left\| \mathbb{E}_P[Y] \right\| \tag{4.1}$$

[1] For simplicity, we choose $\mathcal{Y}$ and $\mathcal{A}$ to have the same size (diameter 2). This is just for exposition, and all results in this section should be easily extensible to $\mathcal{A}$ of different diameter, with appropriate scaling of Lipschitz constants and comparators ϕ.

We first note that $L(P)$ is not differentiable when P has mean 0, but is identical for all such P. Recalling Proposition 6, we choose $\mathcal{P} = \mathcal{P}^{(0)} = \{P \in \Delta(\mathcal{B}) : \mathbb{E}_P[Y] = 0\}$.

We'll show that this restriction (to the family of bounded martingale difference sequences), yields a rich class of stable ϕ. And, under $\mathcal{P}$, Theorem 5 allows us to characterize the minimax value of OLO games in terms of expected values of functions of martingales.

4.2 Sufficient Condition For Stability

Lemma 14 If a function $f : \mathcal{B} \to \mathbb{R}$ satisfies the spherical 1-Lipschitz condition:

$$\|y\| = \|y'\| \implies \left|f(y) - f(y')\right| \leq \|y - y'\| \tag{4.2}$$

then f is stable with respect to $\ell(a, y) : \mathcal{B} \times \mathcal{B} \to \mathbb{R} = -\langle a, y\rangle$ and $\mathcal{P}^{(0)}$.

We offer a quick proof sketch but defer the full proof to Appendix C. For any $P \in \Delta(\mathcal{Y})$, we want to find a $P^* \in \mathcal{P}^{(0)}$ such that:

$$-\|\mathbb{E}_P[Y]\| + \mathbb{E}_P\left[f(Y)\right] \leq \mathbb{E}_{P^*}\left[f(Y)\right] \tag{4.3}$$

We can construct such a P^* from P, by reflecting enough probability mass of P across the hyperplane orthogonal to the mean of P (and passing through zero) so that P^* has mean zero. Owing to the spherical 1-Lipschitz condition, the increase of $-\|\mathbb{E}_P[Y]\|$ to $\|\mathbb{E}_{P^*}[Y]\| = 0$ outweighs whatever the decrease in $\mathbb{E}[f(Y)]$ might be. See Fig. 4.1.

Conveniently, the spherical 1-Lipschitz condition is preserved across tail expectations, meaning we only need to evaluate it for ϕ coordinate-wise:

Lemma 15 Suppose $\phi : \mathcal{B}^T \to \mathbb{R}$ is bounded. If, for all t and fixed coordinates $\mathbf{y}_{-t}$, ϕ satisfies our spherical 1-Lipschitz condition:

$$\|y_t\| = \|y'_t\| \implies |\phi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T})| \leq \|y_t - y'_t\| \tag{4.4}$$

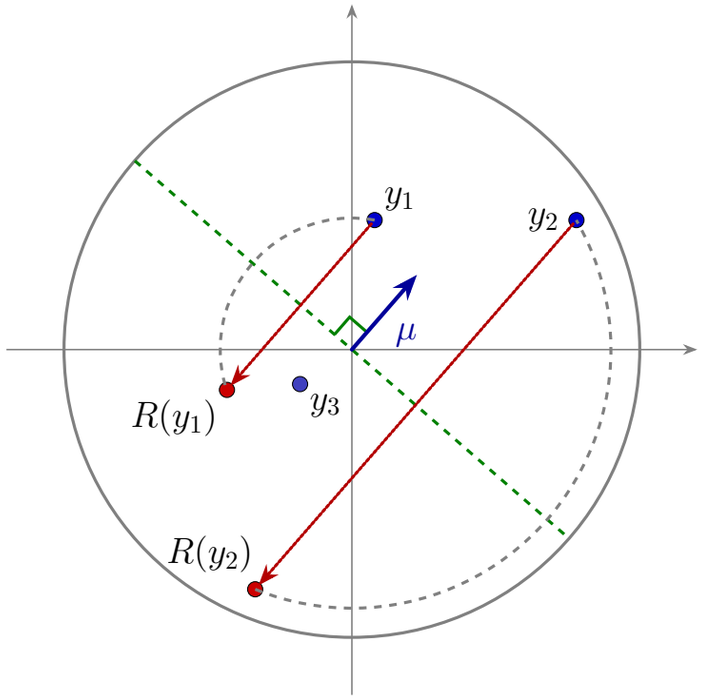


Figure 4.1: Reflection construction for Lemma 14. Distribution P over the 2-dimensional unit ball consists of point masses $y_1, y_2 \in \mathcal{Y}_+$ and $y_3 \in \mathcal{Y}_-$. It has mean μ. To produce P', mass in $\mathcal{Y}_+$ is partially reflected across the hyperplane orthogonal to μ. As mass is reflected from y_1 to $R(y_1)$ (for example), the distance moved by the mass, times the amount of mass moved, is identical to the change in the norm of the mean. And, (as the dashed arcs indicate), $\|R(y_i)\| = \|y_i\|$. So, the spherical 1-Lipschitz condition guarantees that the change in $\mathbb{E}[f(Y)]$ is less than the change from $\|\mu\|$ to 0.

then for all t and all prefixes $\mathbf{y}_{1:t-1}$, $\hat{\phi}_t$ also satisfies the condition:

$$\|y_t\| = \|y'_t\| \implies |\hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y'_t)| \leq \|y_t - y'_t\| \tag{4.5}$$

Proof Recall the definition of $\hat{\phi}_t$:

$$\hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) = \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y_t, Y_{t+1:T})]$$

Suppose $\|y_t\| = \|y'_t\|$ where y_t and y'_t are both in $\mathcal{B}$. WLOG, $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) \geq \hat{\phi}_t(\mathbf{y}_{1:t-1}, y'_t)$.

$$
\begin{aligned}
& |\hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y'_t)| \\
= \; & \hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y'_t) \\
= \; & \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y_t, Y_{t+1:T})] - \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y'_t, Y_{t+1:T})] \\
\leq \; & \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \left(\mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y_t, Y_{t+1:T})] - \mathbb{E}_{P_{t+1:T}}[\phi(\mathbf{y}_{1:t-1}, y'_t, Y_{t+1:T})] \right) \\
= \; & \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}} \left[\phi(\mathbf{y}_{1:t-1}, y_t, Y_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y'_t, Y_{t+1:T}) \right] \\
\leq \; & \sup_{\mathbf{y}_{t+1:T} \in \mathcal{B}^{T-t}} \left(\phi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) \right) \\
\leq \; & \sup_{\mathbf{y}_{t+1:T} \in \mathcal{B}^{T-t}} \left| \phi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) \right|
\end{aligned}
$$

By assumption (4.4), this difference is uniformly bounded above by $\|y_t - y'_t\|$:

$$
\leq \|y_t - y'_t\|
$$

■

Sometimes we can immediately see that ϕ is stable by its structure:

Corollary 16 If ϕ has the form:

$$
\phi(\mathbf{y}_{1:T}) = \psi(\mathbf{y}_{1:T}) + \theta(\|\mathbf{y}_1\|, \ldots, \|\mathbf{y}_T\|) \tag{4.6}
$$

for any ψ which is 1-Lipschitz in each argument, and any bounded $\theta : \mathbb{R}^T \to \mathbb{R}$, then $\hat{\phi}_t$ is stable for all t with respect to the OLO setting, and we can apply Theorem 5.

Once we have confirmed the stability of ϕ, Theorem 5 gives us a stochastic-adversarial

equivalence of the form:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \left(\sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T}) \right) = \sup_{P_{1:T}\in\mathcal{P}_{\mathrm{mds}}} \mathbb{E}_{P_{1:T}} \left[\phi(Y_{1:T}) \right]$$

Where $\mathcal{P}_{\mathrm{mds}}$ is the family of $\mathcal{P}^{(0)}$-stationary distributions (that is, the family of martingale difference sequences.) The right hand-side can be bounded, and occasionally evaluated exactly, by drawing on martingale inequalities.

4.3 Application: External Regret Bound for OLO Over Euclidean Ball

It is easy to show that constrained OLO with standard external regret is stable, and admits a direct stochastic-adversarial equivalence. To make this equivalence as useful as possible, we begin by simplifying the "stochastic" side of the equivalence, before presenting the more useful equality. Throughout, $\mathcal{P} = \mathcal{P}^{(0)}$, the set of zero-mean distributions supported on the unit ball.

Lemma 17

$$\sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\left\| \sum_{t=1}^{T} Y_t \right\| \right] = \sqrt{T}$$

Proof We start by showing that $\sqrt{T}$ upper-bounds the expectation.

$$\begin{aligned}
\sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\left\| \sum_{t=1}^{T} Y_t \right\| \right] &= \sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\sqrt{\left\| \sum_{t=1}^{T} Y_t \right\|^2} \right] \\
&= \sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\sqrt{\sum_{t=1}^{T} \|Y_t\|^2 + \sum_{i\neq j} \langle Y_i, Y_j \rangle} \right]
\end{aligned}$$

By Jensen's inequality,

$$\leq \sup_{P_{1:T}\in\mathcal{P}_{1:T}} \sqrt{\mathbb{E}_{P_{1:T}} \left[\sum_{t=1}^{T} \|Y_t\|^2 + \sum_{i\neq j} \langle Y_i, Y_j \rangle \right]}$$

By the orthogonality and boundedness of the martingale differences,

$$= \sup_{P_{1:T} \in \mathcal{P}_{1:T}} \sqrt{\mathbb{E}_{P_{1:T}}\left[\sum_{t=1}^{T} \|Y_t\|^2\right]} \leq \sqrt{T}$$

For the lower bound, we need to show the existence of a martingale over the unit ball, whose norm after T steps has expected value $\sqrt{T}$. To do so, first consider the last martingale increment Y_T. We'll define P_T to be a uniform distribution over the intersection of the boundary of the ball and the plane orthogonal to the current value of the martingale $\sum_{t=1}^{T-1} Y_t$. Then we have:

$$\mathbb{E}_{P_{1:T}}\left\|\sum_{t=1}^{T} Y_t\right\| = \mathbb{E}_{P_{1:T}}\left[\sqrt{\left\|\sum_{t=1}^{T-1} Y_t\right\|^2 + \|Y_T\|^2 + 2\left\langle \sum_{t=1}^{T-1} Y_t, Y_T\right\rangle}\right] = \mathbb{E}_{P_{1:T-1}}\sqrt{\left\|\sum_{t=1}^{T-1} Y_t\right\|^2 + 1}$$

using the facts that $\|Y_T\| = 1$ and $\langle \sum_{t=1}^{T-1} Y_t, Y_T\rangle = 0$ everywhere in the support of P_T. If we construct Y_{T-1} etc. in the same way, then inductively, we have $\mathbb{E}\left\|\sum_{t=1}^{T} Y_t\right\| = \sqrt{T}$.

■

Proposition 18 For OLO over the Euclidean unit ball with dimension $d \geq 2$:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \left(\sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) - \inf_{a \in \mathcal{B}} \sum_{t=1}^{T} \ell(a, \mathbf{y}_t)\right) = \sqrt{T} \tag{4.7}$$

Proof Define $\phi(\mathbf{y}_{1:T}) = -\inf_{a \in \mathcal{B}} \sum_{t=1}^{T} \ell(a, \mathbf{y}_t) = \left\|\sum_{t=1}^{T} \mathbf{y}_t\right\|$. ϕ is 1-Lipschitz, so we can apply Corollary 16 with $\theta = 0$ and thereby Theorem 5.

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = TL(\mathcal{P}) + \sup_{P_{1:T} \in \mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}}\left[\left\|\sum_{t=1}^{T} Y_t\right\|\right] = \sqrt{T} \tag{4.8}$$

Where we used $L(\mathcal{P}) = 0$, following the logic of eqn. (4.1), and applied Lemma 17. ■

This is only a new result for $d = 2$. [2] show that minimax regret is upper bounded by $\sqrt{T}$. They also give (Lemma 8) a constructive strategy for an adversarial Nature to force exactly $\sqrt{T}$

regret, by choosing each $\mathbf{y}_t$ on the boundary of the ball to be orthogonal to both $\text{Alg}(\mathbf{y}_{1:t-1})$ and $\sum_{t'=1}^{t-1} \mathbf{y}_{t'}$, which is always possible if the dimension is at least 3. Note the similarity between this optimal strategy and the martingale which certifies our equality in the proof of Lemma 17. However, in our martingale approach, we don't need to explicitly consider the learner's behavior at all (and thus can give a guarantee in 2 dimensions).

4.4 Application: Adaptive Regret Bound for OLO Over Euclidean Ball

We start with a purely stochastic martingale result which will be useful for establishing lower bounds on the game value. As previously, $\mathcal{P} = \mathcal{P}^{(0)}$ throughout.

Lemma 19 For $C \leq 2$ and $\mathcal{P} = \mathcal{P}^{(0)}$ (the family of zero-mean distributions over the Euclidean unit ball):

$$\lim_{T\to\infty} \left(\sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\left\| \sum_{t=1}^{T} Y_t \right\| - C\sqrt{\sum_{t=1}^{T} \|Y_t\|^2} \right] \right) \geq \frac{(2-C)^2}{8} \tag{4.9}$$

The full proof is in Appendix C. At a high level, it works by constructing a highly skewed i.i.d. Bernoulli process with zero mean for each time horizon, roughly a constrained version of the construction used in [15]:[2]

$$Q(T) = \begin{cases} \mathbf{u} & \text{w.p.} \quad (2-C)/(4T) \\ -((2-C)/(4T+C-2))\mathbf{u} & \text{w.p.} \quad (4T+C-2)/(4T) \end{cases}$$

where $\mathbf{u}$ is an arbitrary unit vector. Because $Q(T)$ is a Bernoulli distribution, we can find expressions for the expected values of $\|\sum Y_t\|$ and $\sum \|Y_t\|^2$ in terms of a binomially distributed random variable, which approaches a Poisson random variable in the limit, allowing us to use nice linear bounds on the exponential functions showing up in the Poisson PDF.

[2] Rather than proving this entirely new lemma, it might be possible to reach the same conclusion by adapting the proofs of the lower bounds in [15] and [16], to work for martingales with *bounded* differences. This could have the benefit of giving a tighter, T-dependent constant C, allowing us to drop the limits in our claims. However, the original proofs of these lower bounds deeply exploit the unboundedness of the domain and so would require substantial adjustments. (We can't simply replace their construction with our constrained construction, despite the similarities between them).

Using this lemma, we can characterize the value of learning games where the comparator term adapts to the scale of the inputs, as in the regret guarantees given by the AdaGrad algorithm (e.g. eqn. (6) of [5]):

Proposition 20 For OLO over the Euclidean unit ball,

$$\lim_{T\to\infty}\left(\inf_{\mathrm{Alg}_T}\sup_{\mathbf{y}\in\mathcal{Y}^T}\left(\sum_{t=1}^T \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}),\mathbf{y}_t) - \inf_{a\in\mathcal{B}}\sum_{t=1}^T \ell(a,\mathbf{y}_t) - C\sqrt{\sum_{t=1}^T \|\mathbf{y}_t\|^2}\right)\right) \tag{4.10}$$

is exactly 0 for $C = 2$ and greater than 0 for any $C < 2$.

Proof Define

$$\phi(\mathbf{y}_{1:T}) = -\inf_{a\in\mathcal{B}}\sum_{t=1}^T \ell(a,\mathbf{y}_t) - C\sqrt{\sum_{t=1}^T \|\mathbf{y}_t\|^2} = \left\|\sum_{t=1}^T \mathbf{y}_t\right\| - C\sqrt{\sum_{t=1}^T \|\mathbf{y}_t\|^2}$$

Here, we can decompose $\phi = \psi + \theta$ where $\psi(\mathbf{y}) = \left\|\sum_{t=1}^T \mathbf{y}_t\right\|$ is 1-Lipschitz, and $\theta(\|\mathbf{y}_1\|,\dots,\|\mathbf{y}_T\|) = -C\sqrt{\sum_{t=1}^T \|\mathbf{y}_t\|^2}$, so again we can apply Corollary 16 and thereby Theorem 5:

$$\inf_{\mathrm{Alg}_T}\sup_{\mathbf{y}\in\mathcal{Y}^T}\sum_{t=1}^T \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}),\mathbf{y}_t) + \phi(\mathbf{y}) = \cancel{TL(\mathcal{P})} + \sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}}\left[\left\|\sum_{t=1}^T Y_t\right\| - C\sqrt{\sum_{t=1}^T \|Y_t\|^2}\right]$$

One direction of the conclusion then follows from [15], namely the martingale moment inequality: $\mathbb{E}\left[\left\|\sum_t Y_t\right\|\right] \le 2\mathbb{E}\left[\sqrt{\sum_t \|Y_t\|^2}\right]$. Lemma 19 then shows that $C = 2$ is the best possible constant to guarantee the game value remains non-positive. ■

[9] (section 6.2) exhibit an algorithm achieving the same bounds, using Burkholder functions adapted to online learning. Our approach can prove that $C = 2$ is optimal, and additionally is arguably simpler. In the Burkholder method, determining the value of the online learning game requires creatively defining useful sufficient statistics which depend on both the outcomes and the learner's behavior, finding an appropriate Burkholder function, and potentially doing further work to show tightness. By contrast, our framework requires checking the stability of ϕ and finding an

appropriate inequality from stochastic processes, which could include martingale inequalities derived from Burkholder functions, or via any other method.

4.5 Other Regret Notions

In this section, we don't look for novel regret bounds or martingale inequalities, but instead examine some common and uncommon regret notions which have the stability property in bounded OLO.

Lemma 21 If:

$$\phi(\mathbf{y}_{1:T}) = \sup_{a_{1:T} \in \mathcal{C}} \sum_{t=1}^{T} \langle a_t, \mathbf{y}_t \rangle \tag{4.11}$$

for any class of comparators $\mathcal{C} \subseteq \mathcal{B}^T$, then $\phi(\mathbf{y}_{1:t-1}, \cdot, \mathbf{y}_{t+1:T})$ is 1-Lipschitz, for any fixed outcomes $\mathbf{y}_{-t}$.

Proof Take arbitrary $y_1, y_2 \in \mathcal{B}$, and fix $\mathbf{y}_{-t}$ for any t. WLOG, let $\phi(\mathbf{y}_{1:t-1}, y_1, \mathbf{y}_{t+1:T}) \geq \phi(\mathbf{y}_{1:t-1}, y_2, \mathbf{y}_{t+1:T})$. Then,

$$\begin{aligned}
&|\phi(\mathbf{y}_{1:t-1}, y_1, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y_2, \mathbf{y}_{t+1:T})| \\
&= \phi(\mathbf{y}_{1:t-1}, y_1, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y_2, \mathbf{y}_{t+1:T}) \\
&= \sup_{a_{1:T} \in \mathcal{C}} \left(\sum_{i \neq t} \langle a_i, \mathbf{y}_i \rangle + \langle a_t, y_1 \rangle \right) - \sup_{a_{1:T} \in \mathcal{C}} \left(\sum_{i \neq t} \langle a_i, \mathbf{y}_i \rangle + \langle a_t, y_2 \rangle \right) \\
&\leq \sup_{a_{1:T} \in \mathcal{C}} \left(\sum_{i \neq t} \langle a_i, \mathbf{y}_i \rangle + \langle a_t, y_1 \rangle - \sum_{i \neq t} \langle a_i, \mathbf{y}_i \rangle - \langle a_t, y_2 \rangle \right) \\
&= \sup_{a_{1:T} \in \mathcal{C}} (\langle a_t, y_1 - y_2 \rangle) \leq \|y_1 - y_2\|
\end{aligned}$$

■

Corollary 22 (Dynamic Regret) The OLO game based on dynamic regret with path-length

constraints admits a stochastic-adversarial equivalence:

$$\begin{aligned}
&\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \left(\sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) - \inf_{a_{1:T}\in\mathcal{B}^T : \sum_{t=1}^{T-1} \|a_{t+1}-a_t\| \le C} \sum_{t=1}^{T} \ell(a_t, \mathbf{y}_t) \right) \\
&= \sup_{P_{1:T}\in\mathcal{P}_{\mathrm{mds}}} \mathbb{E}_{P_{1:T}} \left[\sup_{a_{1:T}\in\mathcal{B}^T : \sum_{t=1}^{T-1} \|a_{t+1}-a_t\| \le C} \sum_{t=1}^{T} \langle a_t, Y_t \rangle \right]
\end{aligned} \tag{4.12}$$

Corollary 23 ("Switching" Regret) The OLO game based on dynamic regret with k switches admits a stochastic-adversarial equivalence:

$$\begin{aligned}
&\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \left(\sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) - \inf_{a_{1:T}\in\mathcal{B}^T : \sum_{t=1}^{T-1} \mathbb{1}\{a_t \neq a_{t+1}\} \le k} \sum_{t=1}^{T} \ell(a_t, \mathbf{y}_t) \right) \\
&= \sup_{P_{1:T}\in\mathcal{P}_{\mathrm{mds}}} \mathbb{E}_{P_{1:T}} \left[\sup_{a_{1:T}\in\mathcal{B}^T : \sum_{t=1}^{T-1} \mathbb{1}\{a_t \neq a_{t+1}\} \le k} \sum_{t=1}^{T} \langle a_t, Y_t \rangle \right]
\end{aligned} \tag{4.13}$$

It's not clear how useful these equivalences are, but they do automatically say *something* novel about the connection between these games and upper bounds on functions of martingale difference sequences.

Another possibility enabled by the spherical 1-Lipschitz condition involves introducing dependencies on the norms of the outcomes, into the comparator sequence:

Corollary 24 ("Stopped" Regret) The OLO game based on "stopped" regret admits a stochastic-adversarial equivalence:

$$\begin{aligned}
&\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \left(\sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) - \inf_{a\in\mathcal{B}} \sum_{t=1}^{T\wedge\tau} \ell(a, \mathbf{y}_t) : \tau = \inf\left\{ t \in [T] : \sum_{i=1}^{t} \|\mathbf{y}_i\| > V \right\} \right) \\
&= \sup_{P_{1:T}\in\mathcal{P}_{\mathrm{mds}}} \mathbb{E}_{P_{1:T}} \left[\sup_{a\in\mathcal{B}} \sum_{t=1}^{T\wedge\tau} \langle a, Y_t \rangle : \tau = \inf\left\{ t \in [T] : \sum_{i=1}^{t} \|Y_i\| > V \right\} \right]
\end{aligned} \tag{4.14}$$

Here, we effectively force the comparator strategy to just begin predicting 0 zero if there is too much cumulative variance in the outcome sequence (effectively, encoding a hard assumption that cumulative variance will be bounded). This is not a standard regret notion, but illustrates some possibilities. Note that this corollary doesn't follow directly from Lemma 21, but requires applying

Lemma 21 separately to each possible joint level-set of $\{\|\mathbf{y}_1\|, \cdots, \|\mathbf{y}_T\|\}$.

4.6 "Non-Regret" Bounds

As [21] discuss in the discrete prediction setting, we can also treat our choice of ϕ as an open-ended modeling problem, in which we choose ϕ to be larger on sequences we care more about (e.g. because we have some a priori belief that those sequences are more likely). In this approach, we need not define ϕ in terms of a comparator class of alternate algorithms.

A simple example of this might be choosing $\phi(\mathbf{y}_{1:T}) = \frac{1}{2}\sum_{t=1}^{T-1}\langle \mathbf{y}_t, \mathbf{y}_{t+1}\rangle$, reflecting an a priori assumption that Nature exhibits some kind of inertia, similar to a kind of dynamic regret. It's easy to verify that this ϕ is coordinatewise 1-Lipschitz, and therefore stable. And, because ϕ is linear, it is also easy to verify that its expected value over any martingale difference sequence is exactly 0. Thus, we can guarantee that there exists a strategy for the learner which guarantees a loss of at most $-\frac{1}{2}\sum_{t=1}^{T-1}\langle \mathbf{y}_t, \mathbf{y}_{t+1}\rangle$. (This strategy is very simple: the learner plays 0 on the first round and then $\frac{1}{2}\mathbf{y}_{t-1}$ on every subsequent round).

We could choose the slightly more complicated and slightly more regret-like

$$\phi(\mathbf{y}_{1:T}) = \frac{1}{3}\left(\left\|\sum_{t=1}^{T}\mathbf{y}_t\right\| + \sum_{t=1}^{T-1}\langle \mathbf{y}_t, \mathbf{y}_{t+1}\rangle\right)$$

encoding a belief that Nature is likely to exhibit some kind of global preference in addition to an inertial tendency. (Note that the leading $\frac{1}{3}$ ensures that ϕ is 1-Lipschitz and therefore stable.) Then, we can apply Theorem 5. Recalling the martingale inequality from Section 4.3, this guarantees the existence of a strategy for the learner which achieves a loss at most

$$\frac{1}{3}\left(\sqrt{T} - \left\|\sum_{t=1}^{T}\mathbf{y}_t\right\| - \sum_{t=1}^{T-1}\langle \mathbf{y}_t, \mathbf{y}_{t+1}\rangle\right)$$

Again, it's not entirely clear how useful these kinds of bounds are, but they are possible.

4.7 Stability as a Constraint on Optimal Learners

Here, we formalize in the OLO setting a principle implicit in [4] and observed in passing by [18] in the binary prediction setting: that the stability of an upper bound on the value of a learning game helps enforce constraints on a minimax optimal learner. The connection turns out to be surprisingly direct. In OLO, stability implies that the learner gains no advantage by being allowed to deviate from the unit ball. That is, stability ensures that the constraints on the learner aren't active, and could be relaxed.

To prove this, we go through a number of intermediate results.

4.7.1 Helpful Results About Real-Valued Functions

To begin, a few quick notes on notation: $f_t(\mathbf{y}_1, \ldots, \mathbf{y}_T)$ refers to f as a function of the t-th argument, holding all other function arguments constant, $\text{cenv}(f(y))$ refers to the concave envelope of f evaluated at point y, and $f^\dagger$ refers to the concave conjugate of f.

We next find an alternate characterization of the supremum over constant-mean distributions, so that we can bring the tools of convex analysis to bear:

Lemma 25 For $f : \mathcal{Y} \to \mathbb{R}$,

$$\text{cenv}(f(y)) = \sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y] = y} \mathbb{E}_P[f(Y)] \tag{4.15}$$

This is unsurprising, but will be useful. The proof can be found in Appendix C.

Our next lemma will be our main convex analysis tool.

Lemma 26 Given bounded-above function $f : \mathcal{Y} \to \mathbb{R}$,

$$\text{cenv}(f(y)) \leq \text{cenv}(f(0)) + \|y\| \tag{4.16}$$

for all $y \in \mathcal{Y}$, if and only if:

$$\sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{Y}} \left(\langle w, y \rangle - f(y) \right) = \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{Y}} \left(\langle w, y \rangle - f(y) \right) \tag{4.17}$$

Proof We first note that $\text{cenv}(f) = f^{\dagger\dagger}$, where $f^{\dagger\dagger}$ denotes the concave biconjugate of f. We'll use $\partial^+ f$ to denote the superdifferential of f. We start with our stability condition (4.16).

$$\begin{aligned}
& \forall y \in \mathcal{Y} : \quad \text{cenv}(f(y)) \leq \text{cenv}(f(0)) + \|y\| \\
\iff \quad & \forall y \in \mathcal{Y} : \quad f^{\dagger\dagger}(y) \leq f^{\dagger\dagger}(0) + \|y\| \\
\iff \quad & \forall y \in \mathcal{Y} : \quad f^{\dagger\dagger}(y) - \|y\| \leq f^{\dagger\dagger}(0) - \|0\|
\end{aligned}$$

Since $f^{\dagger\dagger}(y) - \|y\|$ is concave, by Fermat's rule,

$$\iff \qquad 0 \in \partial^+ \left(f^{\dagger\dagger}(y) - \|y\| \right)(0)$$

Because the domains of $f^{\dagger\dagger}$ and $\|\cdot\|$ have non-empty intersection,

$$\iff \qquad 0 \in \partial^+ f^{\dagger\dagger}(0) + \partial^+ \left(- \|y\| \right)(0)$$

where $+$ denotes the Minkowski sum. Evaluating the superdifferential of $-\|y\|$,

$$\begin{aligned}
\iff \quad & 0 \in \partial^+ f^{\dagger\dagger}(0) + \mathcal{B} \\
\iff \quad & \exists w^* \in \mathcal{B} : \quad w^* \in \partial^+ f^{\dagger\dagger}(0)
\end{aligned}$$

By the equality condition of the Fenchel-Young inequality:

$$\iff \qquad \exists w^* \in \mathcal{B} : \quad 0 \in \partial^+ f^{\dagger}(w^*)$$

By the concavity of $f^\dagger$ and Fermat's rule,

$$\iff \qquad \exists w^* \in \mathcal{B}: \quad w^* \in \operatorname*{argmax}_{w \in \mathbb{R}^d} f^\dagger(w)$$

Because $\mathcal{B}$ is compact and $f^\dagger$ is upper semicontinuous,

$$\iff \qquad \sup_{w \in \mathbb{R}^d} f^\dagger(w) = \sup_{w \in \mathcal{B}} f^\dagger(w)$$

By the definition of concave conjugate,

$$\iff \qquad \sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) = \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y))$$

■

4.7.2 Concave Envelope Characterizes Stability

Lemma 27 For all t, $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$, and $y \in \mathcal{B}$,

$$\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)\right) \leq \operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, 0)\right) + \|y\| \tag{4.18}$$

if and only if $\hat{\phi}_t(\mathbf{y}_{1:t-1}, \cdot)$ is stable for all t and $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$. As a result, if we have eqn. (4.18), then we can apply Theorem 5.

Proof First observe:

$$L(P) = \inf_{a \in \mathcal{A}} \mathbb{E}_P[\ell(a, Y)] = \inf_{a \in \mathcal{A}} \langle a, \mathbb{E}_P[Y] \rangle = \left\langle \frac{-\mathbb{E}_P[Y]}{\|\mathbb{E}_P[Y]\|}, \mathbb{E}_P[Y] \right\rangle = -\|\mathbb{E}_P[Y]\|$$

Now we can break down the left-hand side of our stability condition (2.2):

$$\begin{aligned}
&\sup_{P\in\Delta(\mathcal{Y})}\left(L(P)+\mathbb{E}_P\left[\hat{\phi}_t(\mathbf{y}_{1:t-1},Y)\right]\right)\\
=\ &\sup_{\mu\in\mathcal{B}}\sup_{P\in\Delta(\mathcal{Y}):\mathbb{E}_P[Y]=\mu}\left(L(P)+\mathbb{E}_P\left[\hat{\phi}_t(\mathbf{y}_{1:t-1},Y)\right]\right)\\
=\ &\sup_{\mu\in\mathcal{B}}\left(-\|\mu\|+\sup_{P\in\Delta(\mathcal{Y}):\mathbb{E}_P[Y]=\mu}\mathbb{E}_P\left[\hat{\phi}_t(\mathbf{y}_{1:t-1},Y)\right]\right)
\end{aligned}$$

Applying Lemma 25,

$$=\ \sup_{\mu\in\mathcal{B}}\left(-\|\mu\|+\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},\mu)\right)\right)$$

And the right-hand-side:

$$\sup_{P\in\mathcal{P}}\left(L(P)+\mathbb{E}_P\left[\hat{\phi}_t(\mathbf{y}_{1:t-1},Y)\right]\right)=\sup_{P\in\Delta(Y):\mathbb{E}_P[Y]=0}\left(-\|0\|+\mathbb{E}_P\left[\hat{\phi}_t(\mathbf{y}_{1:t-1},Y)\right]\right)$$

Applying Lemma 25,

$$=\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},0)\right)$$

So, our stability condition is equivalent to:

$$\sup_{\mu\in\mathcal{B}}\left(-\|\mu\|+\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},\mu)\right)\right)=\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},0)\right)$$

Since the right-hand-side is equal to the left evaluated at $\mu=0$, the left hand side is always greater than the right, so the condition becomes

$$\sup_{\mu\in\mathcal{B}}\left(-\|\mu\|+\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},\mu)\right)\right)\leq\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1},0)\right)$$

With a rename of variables, this is clearly equivalent to our optimality condition eqn. (4.18).

$$\forall y \quad \operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)\right) \leq \operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, 0)\right) + \|y\|$$

■

We can finally prove our main claim of the section:

Proposition 28 Given $T \in \mathbb{N}$, and bounded measurable function $\phi : \mathcal{B}^T \to \mathbb{R}$, if $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is stable with respect to $\mathcal{P}^{(0)}$ and $\ell(a, y) = -\langle a, y\rangle$, for all $t \leq T$ and all $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$, then:

$$\begin{aligned} &\inf_{w_1 \in \mathbb{R}^d} \sup_{\mathbf{y}_1 \in \mathcal{B}} \dots \inf_{w_T \in \mathbb{R}^d} \sup_{\mathbf{y}_T \in \mathcal{B}} \left(\sum_{i=1}^{T} -\langle w_i, \mathbf{y}_i\rangle + \phi(\mathbf{y}_{1:T}) \right) \\ &= \inf_{w_1 \in \mathcal{B}} \sup_{\mathbf{y}_1 \in \mathcal{B}} \dots \inf_{w_T \in \mathcal{B}} \sup_{\mathbf{y}_T \in \mathcal{B}} \left(\sum_{i=1}^{T} -\langle w_i, \mathbf{y}_i\rangle + \phi(\mathbf{y}_{1:T}) \right) \end{aligned} \tag{4.19}$$

The proof is fully detailed in Appendix C, but we'll give a sketch here. It proceeds by induction on T. At each step, we separate the analysis of the value of the constrained learning game between the first round and the remaining rounds (a learning game with $T-1$ steps). We then apply Theorem 5 to the remainder to switch between the $\inf\sup\dots$ form of the game value and a form which is an expectation of a comparator term. A combination of Lemma 27 and Lemma 26 allows us to relax the learner's constraint on the first round, and going back through Theorem 5 and the inductive hypothesis gives us the full result.

4.8 Alternate Proof of Stability Condition

Next, we take a short detour to show that the reflection argument from Lemma 14 can be adapted to the dual space to prove the sufficiency of the spherical 1-Lipschitz property for stability.

Lemma 29 If a function $f : \mathcal{Y} \to \mathbb{R}$ satisfies the spherical 1-Lipschitz property:

$$\|y\| = \|y'\| \implies |f(y) - f(y')| \leq \|y - y'\| \tag{4.20}$$

then

$$\sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) = \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) \tag{4.21}$$

The full proof is in Appendix C. Its main advantage over explicitly constructing a probability distribution (as in the proof of Lemma 14) is that its proof doesn't require integrating over a whole probability distribution. Only one reflected point needs to be considered. See Fig. 4.2 for illustration.

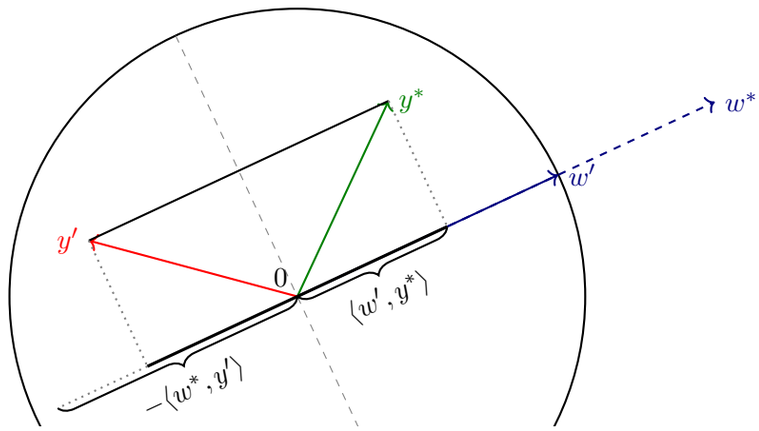


Figure 4.2: The vector w' (on the boundary of the ball) will always outperform w^* (outside the ball). y^* is the best response to w', so if the play (w', y^*) outperforms just *some* other play (w^*, y'), where y' is a not-necessarily-optimal response, then w' will strictly outperform w^*. Choosing y' as in the figure works, since the difference $\langle w', y^* \rangle - \langle w^*, y' \rangle$ is strictly greater than $\|y^* - y'\|$ and therefore $|f(y^*) - f(y')|$.

Recovering the conclusion, that spherical 1-Lipschitz'ness of $\hat{\phi}$ suffices for stability, is then very straightforward:

Corollary 30 If, for all t and fixed coordinates $\mathbf{y}_{-t}$, ϕ_t satisfies our spherical 1-Lipschitz condition:

$$\|y_t\| = \|y_t'\| \implies |\phi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y_t', \mathbf{y}_{t+1:T})| \leq \|y_t - y_t'\| \tag{4.22}$$

then $\hat{\phi}_t$ is stable and we can apply Theorem 5.

Proof Let $t \leq T$, and $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$ be any prefix. By Lemma 15, $\hat{\phi}_t$ satisfies our spherical 1-Lipschitz condition:

$$\|y_t\| = \|y'_t\| \implies |\hat{\phi}_t(\mathbf{y}_{1:t-1}, y_t) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y'_t)| \leq \|y_t - y'_t\|$$

Then, by Lemma 29,

$$\sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{B}} \left(\langle w, y \rangle - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y) \right) = \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{B}} \left(\langle w, y \rangle - \hat{\phi}_t(\mathbf{y}_{1:t-1}, y) \right)$$

And by Lemma 26,

$$\operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)\right) \leq \operatorname{cenv}\left(\hat{\phi}_t(\mathbf{y}_{1:t-1}, 0)\right) + \|y\|$$

Lemma 27 allows us to conclude that $\hat{\phi}_t$ is stable. ■

Chapter 5

Unconstrained Online Linear Optimization

We wind down our results by exploring how stochastic-adversarial equivalences haven't always been fully exploited, even in settings in which they are already known to hold. Specifically, we consider the sub-setting of OLO in which regret benchmarks are scaled according to the norm of the comparator rather than the norms of the outcomes, and in which $\mathcal{A} = \mathbb{R}^d$. The problem setting was initially explored in [13], and is often also called parameter-free learning, as it enables gradient descent algorithms which don't require an initial guess at learning rate based on the scale of the problem.

Given Proposition 28, it should not be surprising that when the learner is unconstrained, stability is not an issue.

Lemma 31 In unconstrained OLO, every bounded-above function f is stable.

Proof First observe, $L(P) = \begin{cases} 0 & P \in \mathcal{P}^{(0)} \\ -\infty & \text{o.w.} \end{cases}$. Say f is bounded above by M. Consider any $P \notin \mathcal{P}^{(0)}$, and let $\delta_0 \in \mathcal{P}^{(0)}$ be the Dirac distribution at 0:

$$L(P) + \mathbb{E}_P[f(Y)] \leq -\infty + M \leq f(0) = L(\delta_0) + \mathbb{E}_{\delta_0}[f(Y)] \leq \sup_{P' \in \mathcal{P}^{(0)}} L(P') + \mathbb{E}_{P'}[f(Y)]$$

So, the supremum over the left-hand side of the inequality is bounded above by the right-hand side, giving us our stability condition. ■

Thus, we can apply Theorem 5 to any regret bound. This conclusion was already established by [12], Corollary 2. That paper, however, did not attempt to apply existing inequalities from the theory of martingales to the unconstrained setting. We will make such an attempt.

5.1 A Regret Bound From Normal Potentials

Lemma 32 Given $T \in \mathbb{N}$, for any regret bound $R : \mathbb{R}_+ \to \mathbb{R}$ based on the norm of the comparator w, such that there exists constant C guaranteeing $R(\|w\|) \geq T\|w\| - C$ for all $w \in \mathbb{R}^d$:

$$\inf_{\text{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sup_{w \in \mathbb{R}^d} \sum_{t=1}^{T} \left(\langle \text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t \rangle - \langle w, \mathbf{y}_t \rangle \right) - R(\|w\|) = \sup_{P_{1:T} \in \mathcal{P}_{\text{mds}}} \mathbb{E}_{P_{1:T}} \left[R^* \left(\left\| \sum_{i=1}^{T} Y_i \right\| \right) \right] \quad (5.1)$$

where R^* is the convex conjugate of $R_{\text{ext}}(x) = \begin{cases} R(x) & x \geq 0 \\ \infty & x < 0 \end{cases}$.

Proof We begin by simplifying ϕ:

$$\begin{aligned} \phi(\mathbf{y}_{1:T}) &= \sup_{w \in \mathbb{R}^d} \left(\left(\sum_{t=1}^{T} -\langle w, \mathbf{y}_t \rangle \right) - R(\|w\|) \right) \\ &= \sup_{w \in \mathbb{R}^d} \left(\left\langle w, -\sum_{t=1}^{T} \mathbf{y}_t \right\rangle - R(\|w\|) \right) = R^* \left(\left\| \sum_{t=1}^{T} \mathbf{y}_t \right\| \right) \end{aligned}$$

As in the proof of Proposition 2 from [9], we used the standard fact from convex analysis that $(f(\|\cdot\|))^* = f^*(\|\cdot\|_*)$ for general dual norm pairs, taking $f = R_{\text{ext}}$. Before we can apply our theorem, we first must confirm that ϕ is bounded. Consider $R^*(y)$ for $y \in [0, T]$.

$$\begin{aligned} R^*(y) = \sup_{x \in \mathbb{R}}(xy - R_{\text{ext}}(x)) &= \max\left(\sup_{x<0}(xy - R_{\text{ext}}(x)), \sup_{x \geq 0}(xy - R_{\text{ext}}(x)) \right) \\ &= \max\left(-\infty, \sup_{x \geq 0}(xy - R(x)) \right) \quad (5.1) \\ &\leq \sup_{x \geq 0}(xy - (Tx - C)) \\ &= \sup_{x \geq 0}\left(x(y - T)\right) + C = C \end{aligned}$$

Where we used the assumption that $y \leq T$ in the last step. This assumption is valid in our case, which we can see by applying the triangle inequality to $\left\|\sum_{t=1}^{T} \mathbf{y}_t\right\|$, since $\mathbf{y}_t \in \mathcal{B}$ for all t.

By Lemma 31, ϕ is stable, and we can apply Theorem 5 to reach our conclusion. ■

We can characterize valid such R^* using worst-case martingale tail bounds from [17].

Lemma 33 Given a regret bound $R : \mathbb{R}_+ \to \mathbb{R}$ which is strictly convex, differentiable, and monotonically increasing,

$$\sup_{P_{1:T} \in \mathcal{P}_{\text{mds}}} \mathbb{E}_{P_{1:T}}\left[R^*\left(\left\|\sum_{i=1}^{T} Y_i\right\|\right)\right] \leq -R(0) + 2\int_0^\infty \left(R'\right)^{-1}(\alpha) \exp\left(\frac{-\alpha^2}{2T}\right) d\alpha \tag{5.2}$$

Proof Define a shifted function $R_+ = R_{\text{ext}} - R(0)$.

We will first show that R_+^* is monotonically increasing. Suppose $y_1 \geq y_2$. By the same argument as in eqn. (5.1), we can restrict the supremum in the definition of the conjugate to be just over nonnegative values, giving:

$$R_+^*(y_1) = \sup_{x \in \mathbb{R}_+} xy_1 - R_+(x) \geq \sup_{x \in \mathbb{R}_+} xy_2 - R_+(x) = R_+^*(y_2)$$

And because R is non-decreasing,

$$R^*(0) = \sup_{x \in \mathbb{R}_+} -R(x) = -\inf_{x \in \mathbb{R}_+} R(x) = -R(0)$$

Next, we relate R^* and R_+^*:

$$\begin{aligned} R_+^*(y) &= \sup_{x \in \mathbb{R}_+} xy - R_+(x) = \sup_{x \in \mathbb{R}_+} xy - (R(x) - R(0)) \\ &= R(0) + \sup_{x \in \mathbb{R}_+} xy - R(x) = R(0) + R^*(y) = -R^*(0) + R^*(y) \end{aligned}$$

Since $R_+^*(y)$ is monotonically increasing and is 0 at $y = 0$, it is nonnegative over nonnegative domains.

Finally, because R is strictly convex, we can apply Theorem 26.3 from [25] to show that R^* is also differentiable.

We return to the main question:

$$\begin{aligned}\sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \mathbb{E}_{P_{1:T}}\left[R^*\left(\left\|\sum_{i=1}^T Y_i\right\|\right)\right] &= \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_{\mathcal{Y}^T} R^*\left(\left\|\sum_{i=1}^T Y_i\right\|\right)\, dP_{1:T} \\ &= \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_{\mathcal{Y}^T} \left(R_+^*\left(\left\|\sum_{i=1}^T Y_i\right\|\right) - R(0)\right)\, dP_{1:T}\end{aligned}$$

Because P will always be a probability measure,

$$= -R(0) + \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_{\mathcal{Y}^T} R_+^*\left(\left\|\sum_{i=1}^T Y_i\right\|\right)\, dP_{1:T}$$

Using the layer cake representation of the integral (which is valid because R_+^* is nonnegative):

$$= -R(0) + \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_0^\infty \mathbb{P}_{P_{1:T}}\left[R_+^*\left(\left\|\sum_{i=1}^T Y_i\right\|\right) > u\right]\, du$$

By the monotonicity of R^*, it has an inverse:

$$= -R(0) + \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_0^\infty \mathbb{P}_{P_{1:T}}\left[\left\|\sum_{i=1}^T Y_i\right\| > (R^*)^{-1}(u)\right]\, du$$

Using the differentiability of R^*, we substitute $\alpha = (R^*)^{-1}(u)$, so that $\frac{du}{d\alpha} = (R^*)'(\alpha)$

$$\begin{aligned}&= -R(0) + \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_0^\infty \mathbb{P}_{P_{1:T}}\left[\left\|\sum_{i=1}^T Y_i\right\| > \alpha\right]\, (R^*)'(\alpha)\, d\alpha \\ &\le -R(0) + \sup_{P_{1:T}\in\mathcal{P}_{\text{mds}}} \int_0^\infty \mathbb{P}_{P_{1:T}}\left[\sup_{t\in[T]}\left\|\sum_{i=1}^t Y_i\right\| \ge \alpha\right]\, (R^*)'(\alpha)\, d\alpha\end{aligned}$$

Using Pinelis's bound from [17]:

$$\le -R(0) + \int_0^\infty 2\exp\left(\frac{-\alpha^2}{2T}\right)\, (R^*)'(\alpha)\, d\alpha$$

Because R is strictly convex and differentiable, we can treat R^* as a Legendre transform (see Theorem 26.5 from [25]), yielding:

$$= -R(0) + \int_0^\infty 2\exp\left(\frac{-\alpha^2}{2T}\right)(R')^{-1}(\alpha)\,d\alpha$$

■

This lemma provides a new perspective on the choice of potential functions R^* that are in the family $b\exp\left(\|y\|^2/(2cT)\right)$, as in [14] and [9]: a good "Pinelis potential" function should grow as fast as possible to give tight bounds, while still having a derivative which is $o((1/\|y\|)\exp(\|y\|^2/(2T)))$, to guarantee convergence in our tail bound integral.

Proposition 34 An algorithm exists in the unconstrained online linear optimization setting guaranteeing the learner a comparator-dependent regret of at most:

$$R(\|w\|) \leq \|w\|\sqrt{2cT\log\left(\frac{\sqrt{cT}\|w\|}{b}+1\right)} + \frac{2b}{c-1} \tag{5.3}$$

for any constants $c \geq 1$, $b > 0$.

Proof We'll consider roughly the same family of potentials as [14], given by:

$$R^*(\alpha) = b\exp\left(\frac{\alpha^2}{2cT}\right) + C; \qquad (R^*)'(\alpha) = (R')^{-1}(\alpha) = \left(\frac{b}{cT}\right)\alpha\exp\left(\frac{\alpha^2}{2cT}\right)$$

for constants $c > 1, b > 0$.

The integral in our bound becomes:

$$\int_0^\infty (R')^{-1}(\alpha)\exp\left(\frac{-\alpha^2}{2T}\right)d\alpha = \int_0^\infty \left(\frac{b}{cT}\right)\alpha\exp\left(\alpha^2\left(\frac{1-c}{2cT}\right)\right)d\alpha$$

Substituting $u = \alpha^2(c-1)/(2cT)$, for which $du = \alpha(c-1)/(cT)\, d\alpha$

$$
\begin{aligned}
&= \int_0^\infty \left(\frac{b}{cT}\right) \frac{cT}{c-1} \exp\left(-u\right)\, du \\
&= \frac{b}{c-1}
\end{aligned}
$$

So, from Lemma 33, to get a valid regret bound, we want $R(0) = \frac{2b}{c-1}$, or $C = -b - \frac{2b}{c-1}$.

We can apply Lemma 14 from [14] to the offset potential $R^* - C$ to get:

$$
\begin{aligned}
& & (R^* - C)^*(x) &\leq x\sqrt{2cT \log\left(\frac{\sqrt{cT}x}{b} + 1\right)} - b \\
\iff & & (R^*)^*(x) + C &\leq x\sqrt{2cT \log\left(\frac{\sqrt{cT}x}{b} + 1\right)} - b \\
\iff & & (R^*)^*(x) &\leq x\sqrt{2cT \log\left(\frac{\sqrt{cT}x}{b} + 1\right)} + \frac{2b}{c-1}
\end{aligned}
$$

so $R(x) = x\sqrt{2cT \log\left(\frac{\sqrt{cT}x}{b} + 1\right)} + \frac{2b}{c-1}$ is a valid regret bound. ■

This is a slightly different form of bound than in [14]. It particularly has the advantage of allowing the parameter c to get arbitrarily close to 1. Setting $b = (c-1)k\sqrt{T}/2$ gives the $O(\sqrt{T})$ expression:

$$
R(x) \leq x\sqrt{2cT \log\left(\frac{2\sqrt{c}x}{(c-1)k} + 1\right)} + k\sqrt{T} \tag{5.4}
$$

Note that this is still worse than the anytime bounds found in [26], which don't require a tradeoff involving the leading constant. Despite failing to recover an optimal bound, though, our simple application of martingale tail bounds gives us non-trivial regret bounds in the unconstrained setting.

Chapter 6

Conclusion

6.1 Connection to Betting Games

In this section, we'll interpret the entire preceding thesis in gambling terms. This will hopefully help us solidify our intuition for the connections between our concrete problem settings.

Both Cover's paper [4], and much of the literature on unconstrained online linear optimization, include some discussion of betting games.

Cover interprets the binary prediction setting as a betting game, where a prediction p corresponds to a bet of size $p - \frac{1}{2}$ on the outcome of 1 (recall that our report space $\mathcal{A}$ consists of random or probabilistic predictions in $[0, 1]$). Positive bets correspond to winning the bet size if the outcome is 1 and losing the bet size if the outcome is 0, and negative bets correspond to the opposite. Cover formally makes the mathematical argument that with unlimited bet sizes, it's possible to design a betting strategy that guarantees any desired data-dependent risk profile (e.g. winning a large amount on a sequence of all 1's and losing a small amount on everything else), so long as the risk profile wins zero in expectation. He additionally shows that there is a one-to-one correspondence between risk profiles and betting strategies, so that if we have restricted bets (as in our prediction setting, since our bets lie in $[-\frac{1}{2}, \frac{1}{2}]$), we must require that the betting strategy yields allowable bets. This requirement turns out to precisely be the stability condition of eqn. (1.2).

Unconstrained OLO also can also be viewed as a betting game, in which the reward (or negative loss) is the learner/gambler's winnings on each round. [12] show that the achievability of a given benchmark or risk profile is given by the supremum over expected values over martingale

difference sequences.

Cover's key insight in [4] was that achieving a certain risk profile required a certain betting strategy, and so we can figure out whether a risk profile is achievable by looking at whether the betting strategy is allowable by the problem constraints. In unconstrained OLO, the achievability of a given risk profile is not as simple as needing to have expected value 0 over a simple uniform distribution – instead, it requires that the supremum over expected values over martingale difference sequences is 0. However, the story around applying constraints closely mirrors Cover's paper. We require a stability condition which implies that an optimal betting strategy for achieving the risk profile will stay within the required constraints. This is exactly what we argue in Proposition 28.

We can give this a general interpretation via the framework of "gamble spaces," introduced in [10]. In this framework, online learning predictions are viewed as gambles (essentially, choices of functions mapping outcomes to rewards) on outcomes chosen adversarially. [10] shows that this framework can be used to interpret statements in the field of game-theoretic probability, and that there are frequently equivalences between theorems in game-theoretic and measure-theoretic probability. In this light, our stochastic-adversarial equivalences in online learning can be translated into equivalences between game-theoretic and measure-theoretic probability. In the notation of [10], ϕ can be considered a gamble space variable, and our Theorem 5 establishes a general condition on the underlying gamble space, under which $\mathbf{E}[\phi] = \overline{\mathbb{E}}^*[\phi]$. The upshot is that many claims which require unbounded gambles ("upward-scalable", in the more precise terminology of [10]) can be relaxed to include gambles which are bounded but stable.

6.2 Future Work

Here, we present a couple of future directions from this thesis.

6.2.1 Constructive Algorithms

Although our approach is not directly constructive, the Relax-and-Randomize framework of [24] allows one to go from characterizations of the value of a learning game to algorithms. Future

work could look for conditions under which the algorithm generated by a relaxation approach has an efficient implementation, and in particular when (a variant on) the random rollout approach of [24] might apply.

6.2.2 Only-If Direction

One minor extension of our framework from chapter 2 would include establishing conditions under which the reverse direction of Theorem 5 hold. That is, when does the stochastic-adversarial equivalence imply stability under every prefix? We expect that this should follow easily when every element of $\mathcal{P}$ has full support, or if we simply don't require stability for prefixes which have no support under an optimal play by Nature.

6.3 Discussion

In [4], Cover showed that (at least in the binary prediction setting under 0-1 loss), there is a direct equivalence between the minimax regret of online learning under adversarial assumptions and under stochastic assumptions.

Our thesis has shown that this phenomenon can be generalized and extended directly into other loss functions and the OLO setting. We also provide a general framework for understanding when the phenomenon applies.

Our direct connection cannot accommodate all interesting problem settings in online learning (e.g. those which depend on internal regret or other learner-behavior-dependent comparators), and is not particularly useful for loss functions with a smooth Bayes risk. Nevertheless, when it applies, it offers a stochastic tool for analyzing the value of a learning game that is much simpler than existing tools like sequential Rademacher complexity or the Burkholder method. It demonstrates that the phenomenon Cover observed is much more than a curious quirk of the binary prediction problem.

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Appendix A

Proofs From Chapter 2

Theorem 5. *We are given $T \in \mathbb{N}$, a measurable, bounded function $\phi : \mathcal{Y}^T \to \mathbb{R}$ which is upper semicontinuous in each coordinate, an ℓ-constant family $\mathcal{P} \subseteq \Delta(\mathcal{Y})$ of conditional distributions, and the $\mathcal{P}$-stationary family $\mathcal{P}_{1:T} \subseteq \Delta(\mathcal{Y}^T)$ of joint distributions.*

If for all $t \leq T$ and $\mathbf{y}_{1:t-1} \in \mathcal{Y}^{t-1}$, $f(y) = \hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is stable with respect to ℓ on $\mathcal{P}$, then

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = TL(\mathcal{P}) + \sup_{P_{1:T} \in \mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}}[\phi(Y_{1:T})] \tag{A.1}$$

where Alg_T is the set of all algorithms in $\mathcal{Y}^{<T} \to \mathcal{A}$.

Proof Proceed by induction on T. For $T = 0$, our claim (2.3) reduces to $\phi(\boldsymbol{\varepsilon}) = \phi(\boldsymbol{\varepsilon})$, which is trivially true. Now assume the statement holds for $T - 1$. Let's start with the left hand side of eq. (2.3).

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) \tag{A.2}$$

$$= \inf_{\mathrm{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \left(\inf_{a \in \mathcal{A}} \sup_{\mathbf{y}_T \in \mathcal{Y}} \ell(a, \mathbf{y}_T) + \phi(\mathbf{y}_{1:T-1}, \mathbf{y}_T) \right) \tag{A.3}$$

$$= \inf_{\mathrm{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \left(\inf_{a \in \mathcal{A}} \sup_{P \in \Delta(\mathcal{Y})} \mathbb{E}_P \left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y) \right] \right) \tag{A.4}$$

Applying Ky Fan's minimax theorem (and here relying on our assumptions as discussed in Section 2.2.1),

$$= \inf_{\text{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \left(\sup_{P \in \Delta(\mathcal{Y})} \inf_{a \in \mathcal{A}} \mathbb{E}_P \left[\ell(a, Y) + \phi(\mathbf{y}_{1:T-1}, Y) \right] \right) \tag{A.5}$$

$$= \inf_{\text{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \left(\sup_{P \in \Delta(\mathcal{Y})} L(P) + \mathbb{E}_P \left[\phi(\mathbf{y}_{1:T-1}, Y) \right] \right) \tag{A.6}$$

and the stability of $\hat{\phi}$, noting that $\hat{\phi}_T = \phi$,

$$= \inf_{\text{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \left(\sup_{P \in \mathcal{P}} L(P) + \mathbb{E}_P \left[\phi(\mathbf{y}_{1:T-1}, Y) \right] \right) \tag{A.7}$$

and the ℓ-constancy of $\mathcal{P}$,

$$= \inf_{\text{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + L(\mathcal{P}) + \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[\phi(\mathbf{y}_{1:T-1}, Y) \right] \tag{A.8}$$

We would like to be able to apply our inductive hypothesis. So, define $\psi(\mathbf{y}_{1:T-1}) : \mathcal{Y}^{T-1} \to \mathbb{R} = \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[\phi(\mathbf{y}_{1:T-1}, Y) \right]$, which we hope to treat as a replacement for ϕ with $T-1$. To evaluate whether ψ meets our stability condition, apply definition 3 to get $\hat{\psi}_t$.

$$\hat{\psi}_t(\mathbf{y}_{1:t-1}, y) \tag{A.9}$$

$$= \sup_{P_{t+1:T-1} \in \mathcal{P}_{t+1:T-1}} \mathbb{E}_{P_{t+1:T-1}} [\psi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T-1})] \tag{A.10}$$

$$= \sup_{P_{t+1:T-1} \in \mathcal{P}_{t+1:T-1}} \mathbb{E}_{P_{t+1:T-1}} \left[\sup_{P_T \in \mathcal{P}} \mathbb{E}_{P_T} \left[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T-1}, Y_T) \right] \right] \tag{A.11}$$

Our inner choice of P_T given a realization of $Y_{t+1:T-1}$ can be equivalently written as a choice over a joint distribution conditioned on $P_{t+1:T-1}$:

$$= \sup_{\substack{P_{t+1:T-1} \in \mathcal{P}_{t+1:T-1} \\ P_T | \mathbf{y}_{1:t-1}, y, \mathbf{y}_{t+1:T-1} \in \mathcal{P} \\ \forall \mathbf{y}_{t+1:T-1} \in \text{supp}(P_{t+1:T-1})}} \mathbb{E}_{P_{t+1:T}} \left[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T}) \right] \tag{A.12}$$

By $\mathcal{P}$-stationarity,

$$= \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}} \left[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T})\right] \tag{A.13}$$

$$= \hat{\phi}_t(\mathbf{y}_{1:t-1}, y) \tag{A.14}$$

Since $\hat{\phi}_t$ is stable for all $t \leq T$, so is $\hat{\psi}_t$ for all $t \leq T-1$. We can therefore apply the induction hypothesis, picking up from eqn. (A.8), giving

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) \tag{A.15}$$

$$= \inf_{\mathrm{Alg}_{T-1}} \sup_{\mathbf{y} \in \mathcal{Y}^{T-1}} \sum_{t=1}^{T-1} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \psi(\mathbf{y}_{1:t-1}) + L(\mathcal{P}) \tag{A.16}$$

$$= (T-1)L(\mathcal{P}) + \sup_{P_{1:T-1} \in \mathcal{P}_{1:T-1}} \mathbb{E}_{P_{1:T-1}}[\psi(Y_{1:T-1})] + L(\mathcal{P}) \tag{A.17}$$

$$= TL(\mathcal{P}) + \sup_{P_{1:T-1} \in \mathcal{P}_{1:T-1}} \mathbb{E}_{P_{1:T-1}} \left[\sup_{P \in \mathcal{P}} \mathbb{E}_P \left[\phi(Y_{1:T-1}, Y)\right]\right] \tag{A.18}$$

Using the same $\mathcal{P}$-stationarity argument as earlier,

$$= TL(\mathcal{P}) + \sup_{P_{1:T} \in \mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\phi(Y_{1:T})\right] \tag{A.19}$$

■

Proposition 7. *Assume $\mathcal{Y}$ is compact, and $\mathcal{P}$ is sequentially closed under the weak-* topology on $\Delta(\mathcal{Y})$. That is, for every sequence of distributions $\{P_n\}_{n=1}^{\infty} \subseteq \mathcal{P}$ satisfying the convergence property:*

$$\lim_{n \to \infty} \mathbb{E}_{P_n} \left[f(Y)\right] = \mathbb{E}_P \left[f(Y)\right] \quad \forall f \in C_b(\mathcal{Y})$$

where $C_b(\mathcal{Y})$ is the set of bounded continuous functions $\mathcal{Y} \to \mathbb{R}$, assume that $P \in \mathcal{P}$. Then, a measurable, bounded, upper semicontinuous (u.s.c.) function $f : \mathcal{Y} \to \mathbb{R}$ is stable with respect to ℓ on

$\mathcal{P}$ if and only if there exists $P^* \in \mathcal{P}$ such that:

$$f \in \partial(-L)(P^*) + N_{\Delta(\mathcal{Y})}(P^*) \tag{A.20}$$

where $+$ is the Minkowski sum, L is the Bayes risk, and $N_{\Delta(\mathcal{Y})}(P^*)$ is the normal cone to the simplex at P^*.

Proof By Theorem 14.22 in [3], $\mathcal{P}$ is compact if and only if it is closed and tight. Because $\mathcal{Y}$ is compact, any subset of $\Delta(\mathcal{Y})$ will be tight, so it suffices to say that $\mathcal{P}$ is closed. And because $\Delta(\mathcal{Y})$ is Polish (because $\mathcal{Y}$ is Polish), it suffices to say that $\mathcal{P}$ is sequentially closed under the weak-* topology. By Theorem 14.3 of [3], our convergence property is equivalent to convergence under the weak-* topology, and our assumption is equivalent to $\mathcal{P}$ being sequentially closed; thus implies that $\mathcal{P}$ is compact.

Consider the right-hand-side of the stability condition: $\sup_{P\in\mathcal{P}} (L(P) + \mathbb{E}_P [f(Y)])$. By Theorem 14.5 of [3], because f is u.s.c., so is the expected value functional $\mathbb{E}_P [f(Y)]$ under the weak-* topology. Similarly, for any fixed $a \in \mathcal{A}$, $\mathbb{E}_P [\ell(a, Y)]$ is u.s.c. in P, by the upper semicontinuity of $\ell(a, \cdot)$. Recall that $L(P) = \inf_{a\in\mathcal{A}} \mathbb{E}_P[\ell(a, Y)]$. A pointwise minimum of u.s.c. functions remains u.s.c., so $L(P)$ is u.s.c. as well.

Because $\mathcal{P}$ is compact, and because the sum of u.s.c. functions is also u.s.c., we can apply the extreme value theorem to guarantee the existence of some $P^* \in \mathcal{P}$ such that:

$$L(P^*) + \mathbb{E}_{P^*} [f(Y)] = \sup_{P\in\mathcal{P}} (L(P) + \mathbb{E}_P [f(Y)])$$

We continue as in the proof of 6, reducing the stability condition to an unconstrained optimality

condition with an indicator function:

$$\begin{aligned}
& \sup_{P\in\Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P\in\mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \\
\iff \quad & \sup_{P\in\Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = L(P^*) + \mathbb{E}_{P^*}\left[f(Y)\right] \\
\iff \quad & \sup_{P\in\mathcal{M}(\mathcal{Y})} \left(L(P) + \langle P, f\rangle - \delta_{\Delta(\mathcal{Y})}(P)\right) = L(P^*) + \langle P^*, f\rangle - \delta_{\Delta(\mathcal{Y})}(P^*) \\
\iff \quad & \forall P' \in M(\mathcal{Y}) \quad L(P') - \delta_{\Delta(\mathcal{Y})}(P') + \left\langle P', f\right\rangle \leq L(P^*) - \delta_{\Delta(\mathcal{Y})}(P^*) + \langle P^*, f\rangle
\end{aligned}$$

which is the definition of the desired subdifferential (recalling the normal cone identity $N_{\Delta(\mathcal{Y})} = \partial\delta_{\Delta(\mathcal{Y})}$),

$$f \in \partial(-L)(P^*) + N_{\Delta(\mathcal{Y})}(P^*)$$

■

Appendix B

Proofs From Chapter 3

Proposition 12. *Under the abstain loss, with the cost of abstention equal to $c \in [0, 1/2]$:*

$$\ell((a,b),y) = ac + (1-a)\left(b\mathbb{1}\{y=0\} + (1-b)\mathbb{1}\{y=1\}\right) \tag{B.1}$$

If, for all prefixes $\mathbf{y}_{1:t-1} \in \{0,1\}^{t-1}$:

$$\mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 1, Y_{t+1:T})\right] - \mathbb{E}_{\mathrm{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, 0, Y_{t+1:T})\right] \in [-1, 0] \tag{B.2}$$

then we can express the exact minimax value of the prediction game over T steps:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T}) = Tc + \mathbb{E}_{\mathrm{Ber}(c)^T}\left[\phi(Y_{1:T})\right] \tag{B.3}$$

In particular, if condition (3.9) holds and $\mathbb{E}_{\mathrm{Ber}(c)^T}\left[-\phi(Y_{1:T})\right] = Tc$, then $-\phi$ represents an achievable data-dependent loss for the learner.

Proof Note:

$$\ell((a,b),y) = ca + (1-a)\left(b\mathbb{1}\{y=0\} + (1-b)\mathbb{1}\{y=1\}\right) \tag{B.4}$$

$$= ca + (1-a)\left(b(1-y) + (1-b)y\right) \tag{B.5}$$

We find the Bayes risk:

$$L(\text{Ber}(p)) = \min_{a\in[0,1],b\in[0,1]} \mathbb{E}_{\text{Ber}(p)}\left[\ell(a,Y)\right] \tag{B.6}$$

$$= \min_{a\in[0,1],b\in[0,1]} \left(ca + (1-a)\left(b(1-p) + (1-b)p\right)\right) \tag{B.7}$$

$$= \min\left(p, 1-p, c\right) \tag{B.8}$$

Take $\mathcal{P} = \{\text{Ber}(c)\}$, and note $L(\text{Ber}(c)) = c$, since $c \leq 1/2$ by assumption. So, our stability condition (from Lemma 9) is:

$$\forall q \in [0,1] \quad L(\text{Ber}(q)) - L(\text{Ber}(p)) \leq (p-q)\left(f(1) - f(0)\right) \tag{B.9}$$

$$\forall q \in [0,1] \quad \min\left(q, 1-q, c\right) - c \leq (c-q)\left(f(1) - f(0)\right) \tag{B.10}$$

We split the three cases $q < c$, $q \in [c, 1-c]$, and $q > 1-c$. In all cases, we divide the inequality by $(c-q)$. This yields the three inequalities:

$$-1 \leq f(1) - f(0) \qquad 0 \geq f(1) - f(0) \qquad \frac{q+1-c}{q-c} \geq f(1) - f(0) \tag{B.11}$$

Combining gives us that stability is equivalent to: $f(1) - f(0) \in [-1, 0]$.

By definition, $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y) = \mathbb{E}_{\text{Ber}(c)^{T-t}}\left[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T})\right]$. Applying Theorem 5, we get:

$$\inf_{\text{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \sum_{t=1}^{T} \ell(\text{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) = TL(\text{Ber}(c)) + \mathbb{E}_{\text{Ber}(c)^T}\left[\phi(Y_{1:T})\right] \tag{B.12}$$

$$= Tc + \mathbb{E}_{\text{Ber}(c)^T}\left[\phi(Y_{1:T})\right] \tag{B.13}$$

■

Lemma 35 If $\mathcal{P} = \{P_1, \ldots, P_k\}$ is a finite set, then a function f is stable with respect to a loss ℓ on $\mathcal{P}$ if and only if there exists some $P_i \in \mathcal{P}$ such that f is stable with respect to ℓ and $\{P_i\}$.

Proof By the finiteness of $\mathcal{P}$, we can reduce the stability condition over $\mathcal{P}$ to:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{B.14}$$

$$= \max_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{B.15}$$

Choose P_{i^*} from the argmax of the right hand side. Then f is stable with respect to $\{P_{i^*}\}$.

Conversely, if f is stable with respect to some $\{P_i\}$, then we have:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P \in \{P_i\}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{B.16}$$

$$\leq \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{B.17}$$

and equality follows because $\mathcal{P} \subseteq \Delta(\mathcal{Y})$. ■

Proposition 36 (Abstain Loss With $\mathcal{P} = \{\mathrm{Ber}(c), \mathrm{Ber}(1-c)\}$) Under the abstain loss, with the cost of abstention equal to $c \in [0, 1/2]$:

$$\ell((a,b),y) = ac + (1-a)\left(b\mathbb{1}\{y=0\} + (1-b)\mathbb{1}\{y=1\}\right) \tag{B.18}$$

If, for all prefixes $\mathbf{y}_{1:t-1} \in \{0,1\}^{t-1}$:

$$\left|\hat{\phi}_t(\mathbf{y}_{1:t-1}, 1) - \hat{\phi}_t(\mathbf{y}_{1:t-1}, 0)\right| \leq 1 \tag{B.19}$$

then we can express the exact minimax value of the prediction game over T steps:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y} \in \mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}_{1:T}) = Tc + \sup_{P_{1:T} \in \{\mathrm{Ber}(c), \mathrm{Ber}(1-c)\}_{1:T}} \mathbb{E}_{P_{1:T}}\left[\phi(Y_{1:T})\right] \tag{B.20}$$

Proof Note:

$$\ell((a,b),y) = ca + (1-a)\left(b\mathbb{1}\{y=0\} + (1-b)\mathbb{1}\{y=1\}\right) \tag{B.21}$$

$$= ca + (1-a)\left(b(1-y) + (1-b)y\right) \tag{B.22}$$

We find the Bayes risk:

$$L(\mathrm{Ber}(p)) = \min_{a\in[0,1],b\in[0,1]} \mathbb{E}_{\mathrm{Ber}(p)}\left[\ell(a,Y)\right] \tag{B.23}$$

$$= \min_{a\in[0,1],b\in[0,1]} \left(ca + (1-a)\left(b(1-p) + (1-b)p\right)\right) \tag{B.24}$$

$$= \min\left(p, 1-p, c\right) \tag{B.25}$$

Take $\mathcal{P} = \{\mathrm{Ber}(c), \mathrm{Ber}(1-c)\}$. Note $L(\mathrm{Ber}(c)) = L(\mathrm{Ber}(1-c)) = c$, since $c \leq 1/2$ by assumption.

We will use Lemma 35 to break down the problem into singleton stability conditions. In the proof of Proposition 12, we derived a stability condition for $\mathcal{P} = \mathrm{Ber}(c)$. For $\mathcal{P} = \mathrm{Ber}(1-c)$, we proceed similarly to that proposition. Our stability condition (from Lemma 9) is:

$$\forall q \in [0,1] \quad L(\mathrm{Ber}(q)) - L(\mathrm{Ber}(p)) \leq (p-q)\left(f(1) - f(0)\right) \tag{B.26}$$

$$\forall q \in [0,1] \quad \min\left(q, 1-q, c\right) - c \leq (1-c-q)\left(f(1) - f(0)\right) \tag{B.27}$$

We split the three cases $q < c$, $q \in [c, 1-c]$, and $q > 1-c$. In all cases, we divide the inequality by $(1-c-q)$. This yields the three inequalities:

$$\frac{q-c}{1-c-q} \leq f(1) - f(0) \qquad 0 \leq f(1) - f(0) \qquad 1 \geq f(1) - f(0) \tag{B.28}$$

Combining gives us that stability is equivalent to: $f(1) - f(0) \in [0,1]$. Using Lemma 35, we combine this with the original stability condition for $\mathcal{P} = \mathrm{Ber}(c)$ and find our stability condition for the larger $\mathcal{P} = \{\mathrm{Ber}(c), \mathrm{Ber}(1-c)\}$: $\|f(1) - f(0)\| \leq 1$, which $\hat{\phi}$ satisfies by assumption. Applying Theorem 5,

we get:

$$\inf_{\mathrm{Alg}_T} \sup_{\mathbf{y}\in\mathcal{Y}^T} \sum_{t=1}^{T} \ell(\mathrm{Alg}(\mathbf{y}_{1:t-1}), \mathbf{y}_t) + \phi(\mathbf{y}) \tag{B.29}$$

$$= TL(\mathcal{P}) + \sup_{P_{1:T}\in\{\mathrm{Ber}(c),\mathrm{Ber}(1-c)\}_{1:T}} \mathbb{E}_{P_{1:T}}\left[\phi(Y_{1:T})\right] \tag{B.30}$$

$$= Tc + \sup_{P_{1:T}\in\{\mathrm{Ber}(c),\mathrm{Ber}(1-c)\}_{T+1:T}} \mathbb{E}_{P_{1:T}}\left[\phi(Y_{1:T})\right] \tag{B.31}$$

■

Appendix C

Proofs From Chapter 4

Lemma 14. *If a function $f : \mathcal{B} \to \mathbb{R}$ satisfies the spherical 1-Lipschitz condition:*

$$\|y\| = \|y'\| \implies \left|f(y) - f(y')\right| \leq \|y - y'\| \tag{C.1}$$

then f is stable with respect to $\ell(a, y) : \mathcal{B} \times \mathcal{B} \to \mathbb{R} = -\langle a, y\rangle$ and $\mathcal{P}^{(0)}$.

Proof Recall our definition of stability:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) = \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{C.2}$$

Suppose for contradiction that:

$$\sup_{P \in \Delta(\mathcal{Y})} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) > \sup_{P \in \mathcal{P}} \left(L(P) + \mathbb{E}_P\left[f(Y)\right]\right) \tag{C.3}$$

(inequality in the other direction is impossible because $\mathcal{P} \subseteq \Delta(\mathcal{Y})$).

Then there exists some $P^* \in \Delta(\mathcal{Y})$ such that for all $P' \in \mathcal{P}$,

$$L(P^*) + \mathbb{E}_{P^*}\left[f(Y)\right] > L(P') + \mathbb{E}_{P'}\left[f(Y)\right] \tag{C.4}$$

$$-\left\|\mathbb{E}_{P^*}[Y]\right\| + \mathbb{E}_{P^*}\left[f(Y)\right] > \mathbb{E}_{P'}\left[f(Y)\right] \tag{C.5}$$

We will construct a particular P' to force a contradiction.

Let $\mu = \mathbb{E}_{P^*}[Y]$ be the mean of P^*, and let $v = \mu/\|\mu\|$ be the normalized direction of the mean. Partition $\mathcal{Y}$ by a hyperplane orthogonal to v:

$$\mathcal{Y}_+ = \{y \in \mathcal{Y} : \langle v, y\rangle > 0\} \tag{C.6}$$

$$\mathcal{Y}_- = \{y \in \mathcal{Y} : \langle v, y\rangle \leq 0\} \tag{C.7}$$

We can decompose the contributions of each partition to $\|\mu\|$:

$$\|\mu\| = \langle \mu, \mu\rangle/\|\mu\| = \langle v, \mu\rangle = \langle v, \mathbb{E}_{P^*}[Y]\rangle \tag{C.8}$$

$$= \mathbb{E}_{P^*}\left[\langle v, Y\rangle\right] \tag{C.9}$$

$$= \int_{\mathcal{Y}_+} \langle v, y\rangle \, dP^*(y) + \int_{\mathcal{Y}_-} \langle v, y\rangle \, dP^*(y) \tag{C.10}$$

Define $M_+ = \int_{\mathcal{Y}_+} \langle v, y\rangle \, dP^*(y)$ to be the contribution from $\mathcal{Y}_+$, and let $\alpha = \|\mu\|/(2M_+)$. Since $\|\mu\| = M_+ + \int_{\mathcal{Y}_-} \langle v, y\rangle \, dP^*(y)$, and $\langle v, y\rangle \leq 0$ for all $y \in \mathcal{Y}_-$ by construction, we know $\|\mu\| \leq M_+$. We also know that $M_+ \geq 0$ by construction of $\mathcal{Y}_+$. So, $\alpha \in [0, 1/2]$.

We will transform P^* into P' via a Markov kernel which reflects points in $\mathcal{Y}_+$ across our partition hyperplane with probability α:

$$\kappa(y, A) = \begin{cases} \delta_y(A) & \text{if } y \in \mathcal{Y}_- \\ (1-\alpha)\delta_y(A) + \alpha\delta_{R(y)}(A) & \text{if } y \in \mathcal{Y}_+ \end{cases} \tag{C.11}$$

where $R(y) = y - 2\langle v, y\rangle v$.

So that,

$$P'(A) = \int_{\mathcal{Y}} \kappa(y, A) dP^*(y) \tag{C.12}$$

$$= P^*(A \cap \mathcal{Y}_-) + (1-\alpha)P^*(A \cap Y_+) + \alpha P^*(R^{-1}(A) \cap Y_+) \tag{C.13}$$

Effectively, this transition moves "enough" mass which is contributing to the non-zero mean to the other side of the outcome space, while preserving the norm, and minimizing the distance moved.

We'll validate that the norm is preserved under $R(y)$:

$$\langle y - 2\langle v, y\rangle v, y - 2\langle v, y\rangle v\rangle \tag{C.14}$$

$$= \langle y, y\rangle - 2\langle v, y\rangle\langle y, v\rangle - 2\langle v, y\rangle\langle v, y\rangle + 4\langle v, y\rangle^2\langle v, y\rangle \tag{C.15}$$

$$= \langle y, y\rangle \tag{C.16}$$

We demonstrate that expected values are affected as intended by our transformation:

$$\mathbb{E}_{P'}[f(Y)] \tag{C.17}$$

$$= \int_{\mathcal{Y}} f(y)\, dP'(y) \tag{C.18}$$

$$= \int_{\mathcal{Y}} \left(\int_{\mathcal{Y}} f(z)\, d\kappa(y, z) \right) dP^*(y) \tag{C.19}$$

$$= \int_{\mathcal{Y}_+} \left(\int_{\mathcal{Y}} f(z)\, d\kappa(y, z) \right) dP^*(y) + \int_{\mathcal{Y}_-} \left(\int_{\mathcal{Y}} f(z)\, d\delta_y(z) \right) dP^*(y) \tag{C.20}$$

$$= \int_{\mathcal{Y}_+} \left(\int_{\mathcal{Y}} f(z)\, d\kappa(y, z) \right) dP^*(y) + \int_{\mathcal{Y}_-} f(y)\, dP^*(y) \tag{C.21}$$

$$= \int_{\mathcal{Y}_+} \left(\int_{\mathcal{Y}} f(z)\, d((1-\alpha)\delta_y(z) + \alpha\delta_{R(y)}(z)) \right) dP^*(y) + \int_{\mathcal{Y}_-} f(y)\, dP^*(y) \tag{C.22}$$

$$= \int_{\mathcal{Y}_+} \left((1-\alpha) \int_{\mathcal{Y}} f(z)\, d\delta_y(z) + \alpha \int_{\mathcal{Y}} f(z)\, d\delta_{R(y)}(z) \right) dP^*(y) + \int_{\mathcal{Y}_-} f(y)\, dP^*(y) \tag{C.23}$$

$$= \int_{\mathcal{Y}_+} (1-\alpha) f(y) + \alpha f(R(y))\, dP^*(y) + \int_{\mathcal{Y}_-} f(y)\, dP^*(y) \tag{C.24}$$

$$= \mathbb{E}_{P^*}[f(Y)] + \alpha \int_{\mathcal{Y}_+} f(R(y)) - f(y)\, dP^*(y) \tag{C.25}$$

We now show that we correctly chose α to force $P' \in \mathcal{P}$:

$$\mathbb{E}_{P'}[Y] = \mathbb{E}_{P^*}[Y] + \alpha \int_{\mathcal{Y}_+} (y - 2\langle v, y\rangle v - y)\ dP^*(y) \tag{C.26}$$

$$= \mu + \alpha \int_{\mathcal{Y}_+} (y - 2\langle v, y\rangle v - y)\ dP^*(y) \tag{C.27}$$

$$= \mu - 2\alpha \int_{\mathcal{Y}_+} \langle v, y\rangle v\, dP^*(y) \tag{C.28}$$

$$= \mu - 2\alpha M_+ v \tag{C.29}$$

$$= \mu - 2\frac{\|\mu\|}{(2M_+)} M_+ \frac{\mu}{\|\mu\|} = 0 \tag{C.30}$$

Now we evaluate the expected value of our spherical 1-Lipschitz function f under P'.

$$\mathbb{E}_{P'}[f(Y)] = \mathbb{E}_{P^*}[f(Y)] + \alpha \int_{\mathcal{Y}_+} (f(R(y)) - f(y))\ dP^*(y) \tag{C.31}$$

$$= \mathbb{E}_{P^*}[f(Y)] - \alpha \int_{\mathcal{Y}_+} (f(y) - f(R(y)))\ dP^*(y) \tag{C.32}$$

$$\geq \mathbb{E}_{P^*}[f(Y)] - \alpha \int_{\mathcal{Y}_+} |f(y) - f(R(y))|\ dP^*(y) \tag{C.33}$$

Because the norm is preserved under our reflection, we can apply our spherical 1-Lipschitz assumption:

$$\geq \mathbb{E}_{P^*}[f(Y)] - \alpha \int_{\mathcal{Y}_+} \|y - (y - 2\langle v, y\rangle v)\|\, dP^*(y) \tag{C.34}$$

$$= \mathbb{E}_{P^*}[f(Y)] - \alpha \int_{\mathcal{Y}_+} \|2\langle v, y\rangle v\|\, dP^*(y) \tag{C.35}$$

$$= \mathbb{E}_{P^*}[f(Y)] - 2\alpha \int_{\mathcal{Y}_+} |\langle v, y\rangle|\, dP^*(y) \tag{C.36}$$

By construction of $\mathcal{Y}_+$, $\langle v, y\rangle > 0$ for all $y \in \mathcal{Y}_+$,

$$= \mathbb{E}_{P^*}[f(Y)] - 2\alpha \int_{\mathcal{Y}_+} \langle v, y\rangle\, dP^*(y) \tag{C.37}$$

$$= \mathbb{E}_{P^*}[f(Y)] - 2\alpha M_+ \tag{C.38}$$

$$= \mathbb{E}_{P^*}[f(Y)] - \|\mu\| \tag{C.39}$$

Contradicting our claim (C.5). ■

Corollary 16. *If ϕ has the form:*

$$\phi(\mathbf{y}_{1:T}) = \psi(\mathbf{y}_{1:T}) + \theta(\|\mathbf{y}_1\|, \ldots, \|\mathbf{y}_T\|) \tag{C.40}$$

for any ψ which is 1-Lipschitz in each argument, and any bounded $\theta : \mathbb{R}^T \to \mathbb{R}$, then $\hat{\phi}_t$ is stable for all t with respect to the OLO setting, and we can apply Theorem 5.

Proof Consider arbitrary $t \leq T$, $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$, $y_t \in \mathcal{B}$, and $y'_t \in \mathcal{B}$, where $\|y_t\| = \|y'_t\|$. Then,

$$\begin{aligned}
&\phi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \phi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) && \text{(C.41)}\\
&= \psi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) + \theta(\|\mathbf{y}_1\|, \ldots, \|y_t\|, \ldots, \|\mathbf{y}_T\|) && \text{(C.42)}\\
&\quad - \psi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) - \theta(\|\mathbf{y}_1\|, \ldots, \|y'_t\|, \ldots, \|\mathbf{y}_T\|) && \text{(C.43)}\\
&= \psi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) + \theta(\|\mathbf{y}_1\|, \ldots, \|y_t\|, \ldots, \|\mathbf{y}_T\|) && \text{(C.44)}\\
&\quad - \psi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) - \theta(\|\mathbf{y}_1\|, \ldots, \|y_t\|, \ldots, \|\mathbf{y}_T\|) && \text{(C.45)}\\
&= \psi(\mathbf{y}_{1:t-1}, y_t, \mathbf{y}_{t+1:T}) - \psi(\mathbf{y}_{1:t-1}, y'_t, \mathbf{y}_{t+1:T}) && \text{(C.46)}\\
&\leq \left\|y_t - y'_t\right\| && \text{(C.47)}
\end{aligned}$$

So by Lemma 15, $\hat{\phi}_t$ is spherical 1-Lipschitz for all t, and by Lemma 14, $\hat{\phi}_t$ is stable for all t. ■

Lemma 19. *For $C \leq 2$ and $\mathcal{P} = \mathcal{P}^{(0)}$ (the family of zero-mean distributions over the Euclidean unit ball):*

$$\lim_{T\to\infty} \left(\sup_{P_{1:T} \in \mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\left\| \sum_{t=1}^{T} Y_t \right\| - C \sqrt{\sum_{t=1}^{T} \|Y_t\|^2} \right] \right) \geq \frac{(2-C)^2}{8} \tag{C.48}$$

Proof Let $\lambda \in [0, 1)$ be a constant whose value will be determined later.

Given fixed $T > \lambda$, let $(Q(T))^T$ be an i.i.d. process with underlying distribution:

$$Q(T) = \begin{cases} \mathbf{u} & \text{w.p.} \quad \frac{\lambda}{T} \\ -\mathbf{u}\frac{\lambda}{T-\lambda} & \text{w.p.} \quad 1 - \frac{\lambda}{T} \end{cases} \tag{C.49}$$

where $\mathbf{u}$ is an arbitrary unit vector, e.g. $(1, 0, 0, \dots)$. It is simple to verify that $(Q(T))^T \in \mathcal{P}_{1:T}$, since $\mathbb{E}_{Q(T)}[Y] = 0$. So,

$$\lim_{T\to\infty} \left(\sup_{P_{1:T}\in\mathcal{P}_{1:T}} \mathbb{E}_{P_{1:T}} \left[\left\| \sum_{t=1}^{T} Y_t \right\| - C\sqrt{\sum_{t=1}^{T} \|Y_t\|^2} \right] \right) \tag{C.50}$$

$$\geq \lim_{T\to\infty} \left(\mathbb{E}_{(Q(T))^T} \left[\left\| \sum_{t=1}^{T} Y_t \right\| - C\sqrt{\sum_{t=1}^{T} \|Y_t\|^2} \right] \right) \tag{C.51}$$

Given a $Y_{1:T}$, define the random variable $X = \sum_{t=1}^{T} \mathbb{1}\left[Y_t = \mathbf{u}\right]$. So,

$$= \lim_{T\to\infty} \left(\mathbb{E}_{(Q(T))^T} \left[\left| X - (T-X)\frac{\lambda}{T-\lambda} \right| - C\sqrt{X + (T-X)\left(\frac{\lambda}{T-\lambda}\right)^2} \right] \right) \tag{C.52}$$

$$= \lim_{T\to\infty} \left(\sum_{k=0}^{T} \mathbb{P}_{(Q(T))^T}\left[X = k\right] \left(\left| k - (T-k)\frac{\lambda}{T-\lambda} \right| - C\sqrt{k + (T-k)\left(\frac{\lambda}{T-\lambda}\right)^2} \right) \right) \tag{C.53}$$

$$= \sum_{k=0}^{\infty} \lim_{T\to\infty} \left(\mathbb{P}_{(Q(T))^T}\left[X = k\right] \left(\left| k - (T-k)\frac{\lambda}{T-\lambda} \right| - C\sqrt{k + (T-k)\left(\frac{\lambda}{T-\lambda}\right)^2} \right) \right) \tag{C.54}$$

$$= \sum_{k=0}^{\infty} \lim_{T\to\infty} \mathbb{P}_{(Q(T))^T}\left[X = k\right] \lim_{T\to\infty} \left(\left| k - (T-k)\frac{\lambda}{T-\lambda} \right| - C\sqrt{k + (T-k)\left(\frac{\lambda}{T-\lambda}\right)^2} \right) \tag{C.55}$$

$$= \sum_{k=0}^{\infty} \lim_{T\to\infty} \mathbb{P}_{(Q(T))^T}\left[X = k\right] \left(|k - \lambda| - C\sqrt{k} \right) \tag{C.56}$$

Where swapping the limit and the sum, and distributing the limit over the products, are justified by the finiteness of the overall expression. Using the Poisson limit theorem,

$$= \sum_{k=0}^{\infty} \exp(-\lambda)\frac{\lambda^k}{k!} \left(|k - \lambda| - C\sqrt{k} \right) \tag{C.57}$$

Because we assumed $\lambda < 1$, we can separate the absolute value cases between $k = 0$ and $k \geq 1$:

$$= \lambda \exp(-\lambda) + \sum_{k=1}^{\infty} \exp(-\lambda) \frac{\lambda^k}{k!} \left(k - \lambda - C\sqrt{k} \right) \tag{C.58}$$

From the AM-GM inequality applied to k and 1:

$$\geq \lambda \exp(-\lambda) + \sum_{k=1}^{\infty} \exp(-\lambda) \frac{\lambda^k}{k!} \left(k - \lambda - C\frac{k+1}{2} \right) \tag{C.59}$$

$$= \lambda \exp(-\lambda) + \sum_{k=0}^{\infty} \exp(-\lambda) \frac{\lambda^k}{k!} \left(k - \lambda - C\frac{k+1}{2} \right) - \left(-\lambda - \frac{C}{2} \right) \exp(-\lambda) \tag{C.60}$$

$$= 2\left(\lambda + \frac{C}{2} \right) \exp(-\lambda) + \sum_{k=0}^{\infty} \exp(-\lambda) \frac{\lambda^k}{k!} \left(k - \lambda - C\frac{k+1}{2} \right) \tag{C.61}$$

$$= 2\left(\lambda + \frac{C}{2} \right) \exp(-\lambda) + \mathbb{E}_{\text{Poisson}(\lambda)} \left[Y - C\frac{Y+1}{2} - \lambda \right] \tag{C.62}$$

Since the mean of the Poisson distribution is just λ,

$$= 2\left(\lambda + \frac{C}{2} \right) \exp(-\lambda) - C\frac{\lambda + 1}{2} \tag{C.63}$$

Using the inequality $\exp(x) \geq 1 + x$,

$$\geq \left(2\lambda + \frac{C}{2} \right) (1 - \lambda) - C\left(\frac{\lambda + 1}{2} \right) \tag{C.64}$$

$$= \lambda(2 - C) - 2\lambda^2 \tag{C.65}$$

To maximize this lower bound, we choose $\lambda = \frac{2-C}{4}$, yielding

$$= \frac{(2-C)^2}{4} - \frac{(2-C)^2}{8} = \frac{(2-C)^2}{8} \tag{C.66}$$

■

Lemma 25. *For $f : \mathcal{Y} \to \mathbb{R}$,*

$$\operatorname{cenv}(f(y)) = \sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y] = y} \mathbb{E}_P[f(Y)] \tag{C.67}$$

Proof We use the characterization of $\operatorname{cenv}(f)$ as the upper bound of the convex hull of the hypograph of f: $\operatorname{hyp}(f) \subseteq \mathcal{Y} \times \mathbb{R} = \{(y, r) : r \leq f(y)\}$. Then

$$\operatorname{cenv}(f(y)) = \sup \{r : (y, r) \in \operatorname{conv}(\operatorname{hyp}(f))\} \tag{C.68}$$

And,

$$\operatorname{conv}(\operatorname{hyp}(f)) = \left\{ \sum_{i=1}^{N} \lambda_i (y_i, r_i) : N \geq 1, r_i \in \mathbb{R}, y_i \in \mathcal{Y}, r_i \leq f(y_i), \sum \lambda_i = 1 \right\} \tag{C.69}$$

(See [25], Section 5).

Define $\Delta_f(\mathcal{Y}) = \left\{ (\lambda_i, y_i)_{i=1}^N : \sum \lambda_i = 1 \right\}$ to be the set of probability distributions over $\mathcal{Y}$ with finite support. So,

$$\operatorname{conv}(\operatorname{hyp}(f)) = \{(\mathbb{E}_P[Y], \mathbb{E}_P[r(Y)]) : r : \mathcal{Y} \to \mathbb{R}, r \leq f, P \in \Delta_f(\mathcal{Y})\} \tag{C.70}$$

and

$$\begin{aligned}
\operatorname{cenv}(f(y)) &= \sup \{r : (y, r) \in \operatorname{conv}(\operatorname{hyp}(f))\} && \text{(C.71)} \\
&= \sup \{\mathbb{E}_P[r(Y)] : r \leq f, P \in \Delta_f(\mathcal{Y}), \mathbb{E}_P[Y] = y\} && \text{(C.72)} \\
&\leq \sup \{\mathbb{E}_P[f(Y)] : P \in \Delta_f(\mathcal{Y}), \mathbb{E}_P[Y] = y\} && \text{(C.73)} \\
&\leq \sup \{\mathbb{E}_P[f(Y)] : P \in \Delta(\mathcal{Y}), \mathbb{E}_P[Y] = y\} && \text{(C.74)} \\
&= \sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y] = y} \mathbb{E}_P[f(Y)] && \text{(C.75)}
\end{aligned}$$

The other direction follows from the fact that $\text{cenv}(f)$ upper bounds f, and Jensen's inequality:

$$\sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y]=y} \mathbb{E}_P[f(Y)] \leq \sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y]=y} \mathbb{E}_P[\text{cenv}(f(Y))] \tag{C.76}$$

$$\leq \sup_{P \in \Delta(\mathcal{Y}): \mathbb{E}_P[Y]=y} \text{cenv}(f(\mathbb{E}_P[Y])) \tag{C.77}$$

$$= \text{cenv}(f(y)) \tag{C.78}$$

■

Proposition 28. *Given $T \in \mathbb{N}$, and bounded measurable function $\phi : \mathcal{B}^T \to \mathbb{R}$, if $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is stable with respect to $\mathcal{P}^{(0)}$ and $\ell(a, y) = -\langle a, y \rangle$, for all $t \leq T$ and all $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$, then:*

$$\begin{aligned} &\inf_{w_1 \in \mathbb{R}^d} \sup_{\mathbf{y}_1 \in \mathcal{B}} \cdots \inf_{w_T \in \mathbb{R}^d} \sup_{\mathbf{y}_T \in \mathcal{B}} \left(\sum_{i=1}^{T} -\langle w_i, \mathbf{y}_i \rangle + \phi(\mathbf{y}_{1:T}) \right) \\ &= \inf_{w_1 \in \mathcal{B}} \sup_{\mathbf{y}_1 \in \mathcal{B}} \cdots \inf_{w_T \in \mathcal{B}} \sup_{\mathbf{y}_T \in \mathcal{B}} \left(\sum_{i=1}^{T} -\langle w_i, \mathbf{y}_i \rangle + \phi(\mathbf{y}_{1:T}) \right) \end{aligned} \tag{C.79}$$

Proof Proceed by induction on T.

The base case $T = 0$ reduces to the claim $\phi(\boldsymbol{\varepsilon}) = \phi(\boldsymbol{\varepsilon})$.

Now assume the claim holds for $T-1$, and that $\hat{\phi}_t(\mathbf{y}_{1:t-1}, y)$ is stable for all $t \leq T$ and prefixes $\mathbf{y}_{1:t-1} \in \mathcal{B}^{t-1}$.

We'll separately consider the first round of the game and the remainder.

To help analysis of the remainder, for any $\mathbf{y}_1$, define $\psi^{(\mathbf{y}_1)}(\mathbf{y}_{2:T}) := \phi(\mathbf{y}_{1:T})$. Note:

$$\hat{\psi}_t^{(\mathbf{y}_1)}(\mathbf{y}_{2:t-1}, y) = \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}} \left[\psi^{(\mathbf{y}_1)}(\mathbf{y}_1, \mathbf{y}_{2:t-1}, y, Y_{t+1:T}) \right] \tag{C.80}$$

$$= \sup_{P_{t+1:T} \in \mathcal{P}_{t+1:T}} \mathbb{E}_{P_{t+1:T}} \left[\phi(\mathbf{y}_{1:t-1}, y, Y_{t+1:T}) \right] \tag{C.81}$$

$$= \hat{\phi}_t(\mathbf{y}_{1:t-1}, y) \tag{C.82}$$

In particular, $\hat{\psi}_t^{(\mathbf{y}_1)}(\mathbf{y}_{2:t-1}, y)$ will be stable for any $\mathbf{y}_1$ and any prefix $\mathbf{y}_{2:t-1}$.

Now we return to the value of the constrained game:

$$\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}\ldots\inf_{w_T\in\mathcal{B}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=1}^{T}-\langle w_i,\mathbf{y}_i\rangle+\phi(\mathbf{y}_{1:T})\right) \tag{C.83}$$

$$=\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\inf_{w_2\in\mathcal{B}}\sup_{\mathbf{y}_2\in\mathcal{B}}\ldots\inf_{w_T\in\mathcal{B}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=2}^{T}-\langle w_i,\mathbf{y}_i\rangle+\phi(\mathbf{y}_{1:T})\right) \tag{C.84}$$

$$=\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\inf_{w_2\in\mathcal{B}}\sup_{\mathbf{y}_2\in\mathcal{B}}\ldots\inf_{w_T\in\mathcal{B}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=2}^{T}-\langle w_i,\mathbf{y}_i\rangle+\psi^{(\mathbf{y}_1)}(\mathbf{y}_{2:T})\right) \tag{C.85}$$

As we observed, $\hat{\psi}_t^{(\mathbf{y}_1)}$ will always be stable, so we can apply Theorem 5 to the remainder:

$$=\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\sup_{P_{2:T}\in\mathcal{P}_{2:T}}\mathbb{E}_{P_{2:T}}\left[\psi^{(\mathbf{y}_1)}(Y_{2:T})\right] \tag{C.86}$$

$$=\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\sup_{P_{2:T}\in\mathcal{P}_{2:T}}\mathbb{E}_{P_{2:T}}\left[\phi(\mathbf{y}_1,Y_{2:T})\right] \tag{C.87}$$

$$=\inf_{w_1\in\mathcal{B}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\hat{\phi}_1(\mathbf{y}_1) \tag{C.88}$$

By assumption, $\hat{\phi}_1$ is stable. Combining Lemma 27 and Lemma 26, we can relax the constraint on w_1:

$$=\inf_{w_1\in\mathbb{R}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\hat{\phi}_1(\mathbf{y}_1) \tag{C.89}$$

Reversing the previous few steps but keeping the relaxed constraint on w_1,

$$=\inf_{w_1\in\mathbb{R}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\inf_{w_2\in\mathcal{B}}\sup_{\mathbf{y}_2\in\mathcal{B}}\ldots\inf_{w_T\in\mathcal{B}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=2}^{T}-\langle w_i,\mathbf{y}_i\rangle+\psi^{(\mathbf{y}_1)}(\mathbf{y}_{2:T})\right) \tag{C.90}$$

Applying the inductive hypothesis,

$$=\inf_{w_1\in\mathbb{R}}\sup_{\mathbf{y}_1\in\mathcal{B}}-\langle w_1,\mathbf{y}_1\rangle+\inf_{w_2\in\mathbb{R}}\sup_{\mathbf{y}_2\in\mathcal{B}}\ldots\inf_{w_T\in\mathbb{R}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=2}^{T}-\langle w_i,\mathbf{y}_i\rangle+\psi^{(\mathbf{y}_1)}(\mathbf{y}_{2:T})\right) \tag{C.91}$$

$$=\inf_{w_1\in\mathbb{R}}\sup_{\mathbf{y}_1\in\mathcal{B}}\ldots\inf_{w_T\in\mathbb{R}}\sup_{\mathbf{y}_T\in\mathcal{B}}\left(\sum_{i=1}^{T}-\langle w_i,\mathbf{y}_i\rangle+\phi(\mathbf{y}_{1:T})\right) \tag{C.92}$$

■

Lemma 29. *If a function $f : \mathcal{Y} \to \mathbb{R}$ satisfies the spherical 1-Lipschitz property:*

$$\|y\| = \|y'\| \implies |f(y) - f(y')| \leq \|y - y'\| \tag{C.93}$$

then

$$\sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) = \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) \tag{C.94}$$

Proof Suppose for contradiction that

$$\sup_{w \in \mathbb{R}^d} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) > \sup_{w \in \mathcal{B}} \inf_{y \in \mathcal{Y}} (\langle w, y \rangle - f(y)) \tag{C.95}$$

so that:

$$\exists w^* \notin \mathcal{B} \ : \ \forall w' \in \mathcal{B} \ : \quad \inf_{y \in \mathcal{Y}} (\langle w^*, y \rangle - f(y)) > \inf_{y \in \mathcal{Y}} \left(\langle w', y \rangle - f(y) \right) \tag{C.96}$$

In particular, this holds if we set $w' = w^*/\|w^*\|$ to be the projection of w^* onto the ball.

We can express condition (C.96) for a particular w^* and $w' = w^*/\|w^*\|$ as:

$$\exists y^* \in \mathcal{Y} \ : \ \forall y' \in \mathcal{Y} \ : \quad \langle w^*, y' \rangle - f(y') > \langle w', y^* \rangle - f(y^*) \tag{C.97}$$

In particular, this holds for $y' = y^*$, so:

$$\langle w^*, y^* \rangle - f(y^*) > \langle w', y^* \rangle - f(y^*) \tag{C.98}$$

$$\langle w^* - w', y^* \rangle > 0 \tag{C.99}$$

$$(\|w^*\| - 1) \langle w', y^* \rangle > 0 \tag{C.100}$$

By assumption, $\|w^*\| > 1$, so,

$$\langle w', y^* \rangle > 0 \tag{C.101}$$

Consider the alternative choice of y' given by the Householder reflection of y^* across the hyperplane orthogonal to w': $y' = y^* - 2\langle w', y^* \rangle w'$.

To verify that this preserves the norm:

$$\begin{aligned}
\langle y', y' \rangle &= \left\langle y^* - 2\langle w', y^* \rangle w', y^* - 2\langle w', y^* \rangle w' \right\rangle && \text{(C.102)} \\
&= \langle y^*, y^* \rangle - 2\langle w', y^* \rangle \langle y^*, w' \rangle - 2\langle w', y^* \rangle \langle y^*, w' \rangle + 4\langle w', y^* \rangle^2 \langle w', w' \rangle && \text{(C.103)} \\
&= \langle y^*, y^* \rangle && \text{(C.104)}
\end{aligned}$$

Note also, because $\|w'\| = 1$:

$$\|y' - y^*\| = 2\langle w', y^* \rangle \tag{C.105}$$

We substitute our new choice of y' into eqn. (C.97):

$$\begin{aligned}
\langle w^*, y' \rangle - f(y') &> \langle w', y^* \rangle - f(y^*) && \text{(C.106)} \\
f(y^*) - f(y') &> \langle w', y^* \rangle - \langle w^*, y' \rangle && \text{(C.107)} \\
&= \langle w', y^* \rangle - \left\langle w^*, y^* - 2\langle w', y^* \rangle w' \right\rangle && \text{(C.108)} \\
&= \langle w', y^* \rangle - \langle w^*, y^* \rangle + 2\langle w', y^* \rangle \langle w^*, w' \rangle && \text{(C.109)} \\
&= \langle w' - w^*, y^* \rangle + 2\langle w', y^* \rangle \|w^*\| && \text{(C.110)} \\
&= (1 - \|w^*\|) \langle w', y^* \rangle + 2\|w^*\| \langle w', y^* \rangle && \text{(C.111)} \\
&= (1 + \|w^*\|) \langle w', y^* \rangle && \text{(C.112)}
\end{aligned}$$

$\|w^*\| > 1$ by assumption, and $\langle w', y^* \rangle > 0$ by eqn. (C.101), so,

$$> 2\langle w', y^* \rangle \tag{C.113}$$

By eqn. (C.105),

$$= \|y' - y^*\| \tag{C.114}$$

contradicting our spherical Lipschitz assumption. ■